US 20250276708A1

(19) **United States**
(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0276708 A1**
**Curran et al.** (43) **Pub. Date: Sep. 4, 2025**

(54) **SYSTEM AND METHOD FOR OPTIMIZATION OF VEHICLE PERCEPTION SENSOR CONFIGURATION**

(71) Applicant: **Toyota Motor Engineering & Manufacturing North America, Inc.**, Plano, TX (US)

(72) Inventors: **Noah T. Curran**, Ann Arbor, MI (US); **Georgios Fainekos**, Novi, MI (US); **Bardh Hoxha**, Canton, MI (US); **Hideki Okamoto**, Ann Arbor, MI (US); **Danil V. Prokhorov**, Canton, MI (US)

(73) Assignees: **Toyota Motor Engineering & Manufacturing North America, Inc.**, Plano, TX (US); **Toyota Jidosha Kabushiki Kaisha**, Aichi-ken (JP)

(21) Appl. No.: **18/593,630**

(22) Filed: **Mar. 1, 2024**

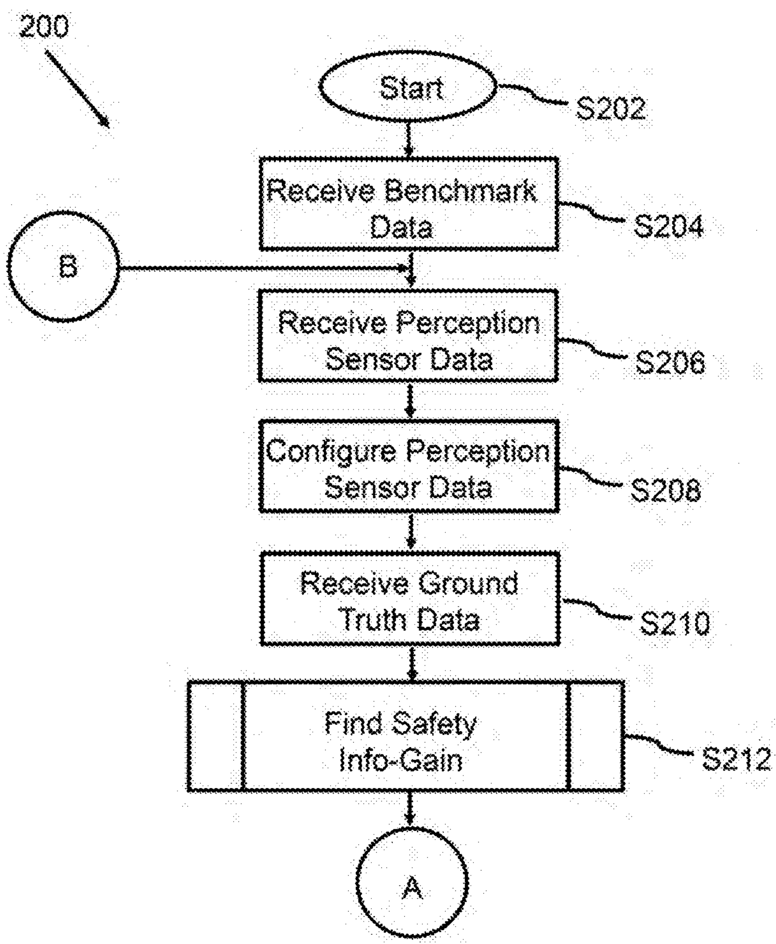
**Publication Classification**

(51) **Int. Cl.**
***B60W 50/14*** (2020.01)
***B60W 60/00*** (2020.01)

(52) **U.S. Cl.**
CPC ........ ***B60W 50/14*** (2013.01); ***B60W 60/0015*** (2020.02); *B60W 2050/146* (2013.01); *B60W 2540/215* (2020.02); *B60W 2554/80* (2020.02)

(57) **ABSTRACT**

A method, computer-readable media, and computer system for receiving perception sensor input data of an autonomous vehicle, the perception sensor input data corresponding to respective measured location, orientation, and type of each of a plurality of perception sensors of the autonomous vehicle; receiving ground-truth information input data of the autonomous vehicle, the ground-truth information input data corresponding to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle; determining, via a processor configured to execute instructions stored in a memory, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and outputting, via the processor, a safety-aware occupancy signal based on the safety occupancy.



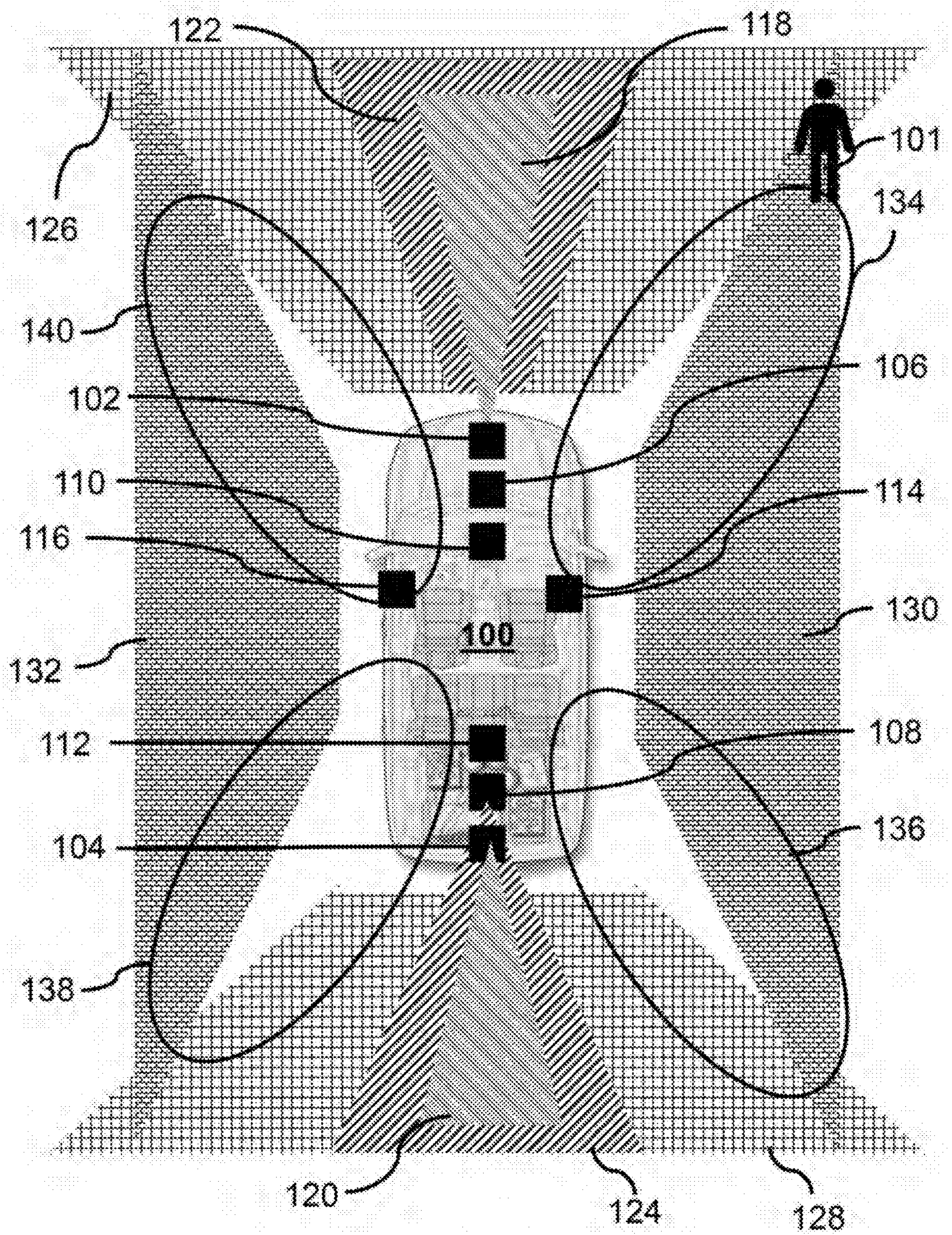

122
118
101
126
134
140
106
102
110
114
116
130
100
132
112
108
136
104
138
120
124
128

FIG. 1A

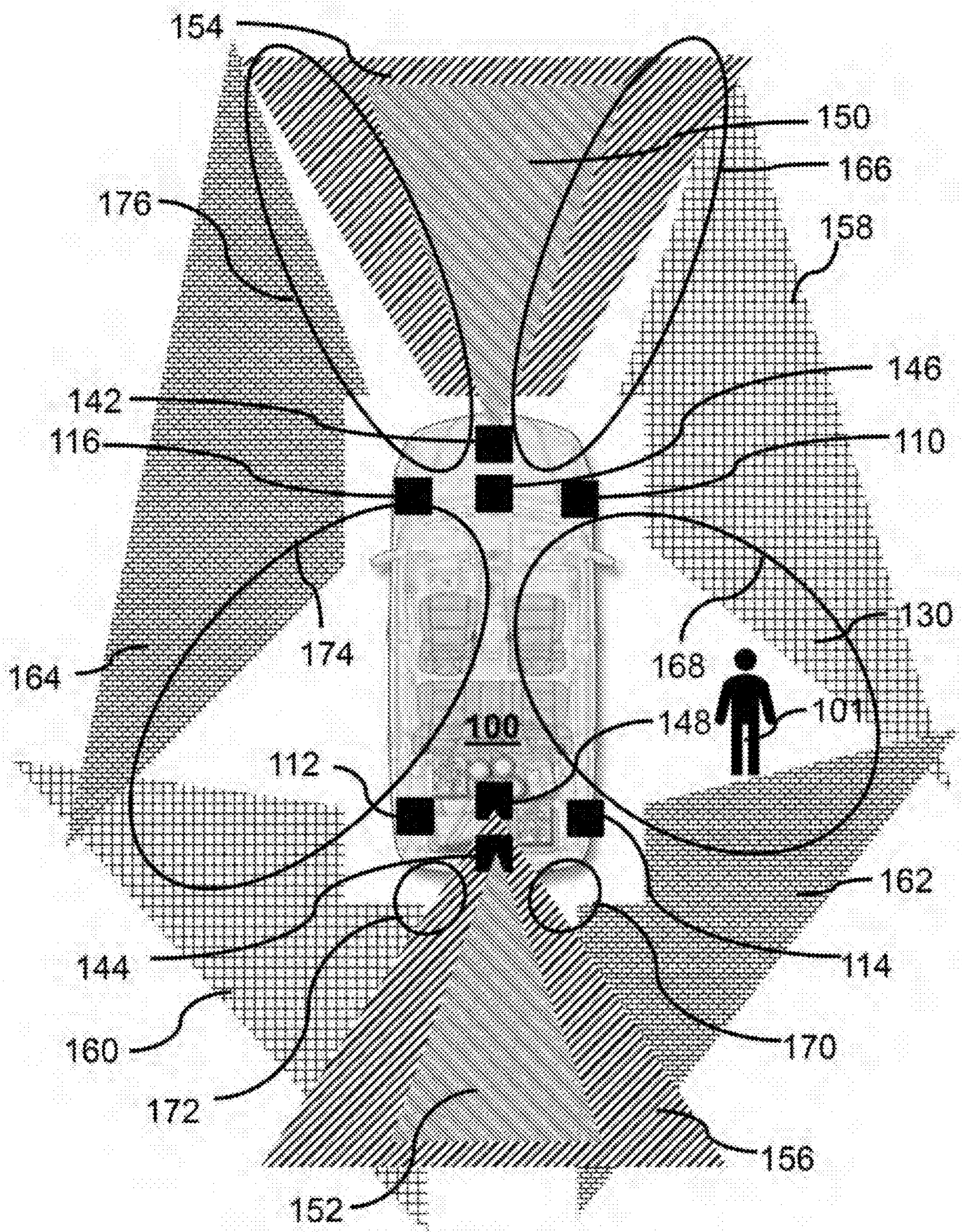

154
150
166
176
158
146
142
116
110
130
168
174
164
101
148
100
112
162
144
114
160
170
172
156
152

FIG. 1B

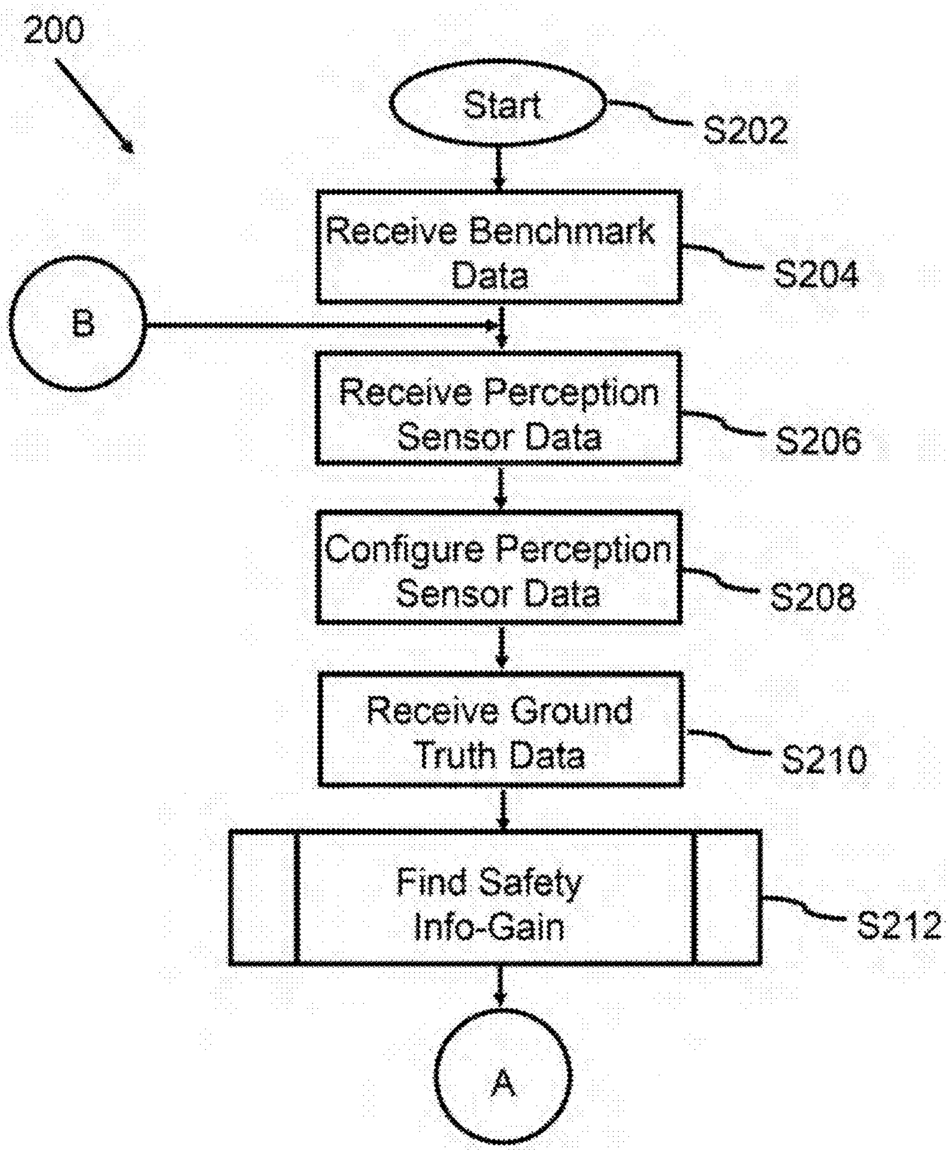

200
Start
S202
Receive Benchmark Data
S204
B
Receive Perception Sensor Data
S206
Configure Perception Sensor Data
S208
Receive Ground Truth Data
S210
Find Safety Info-Gain
S212
A

FIG. 2A

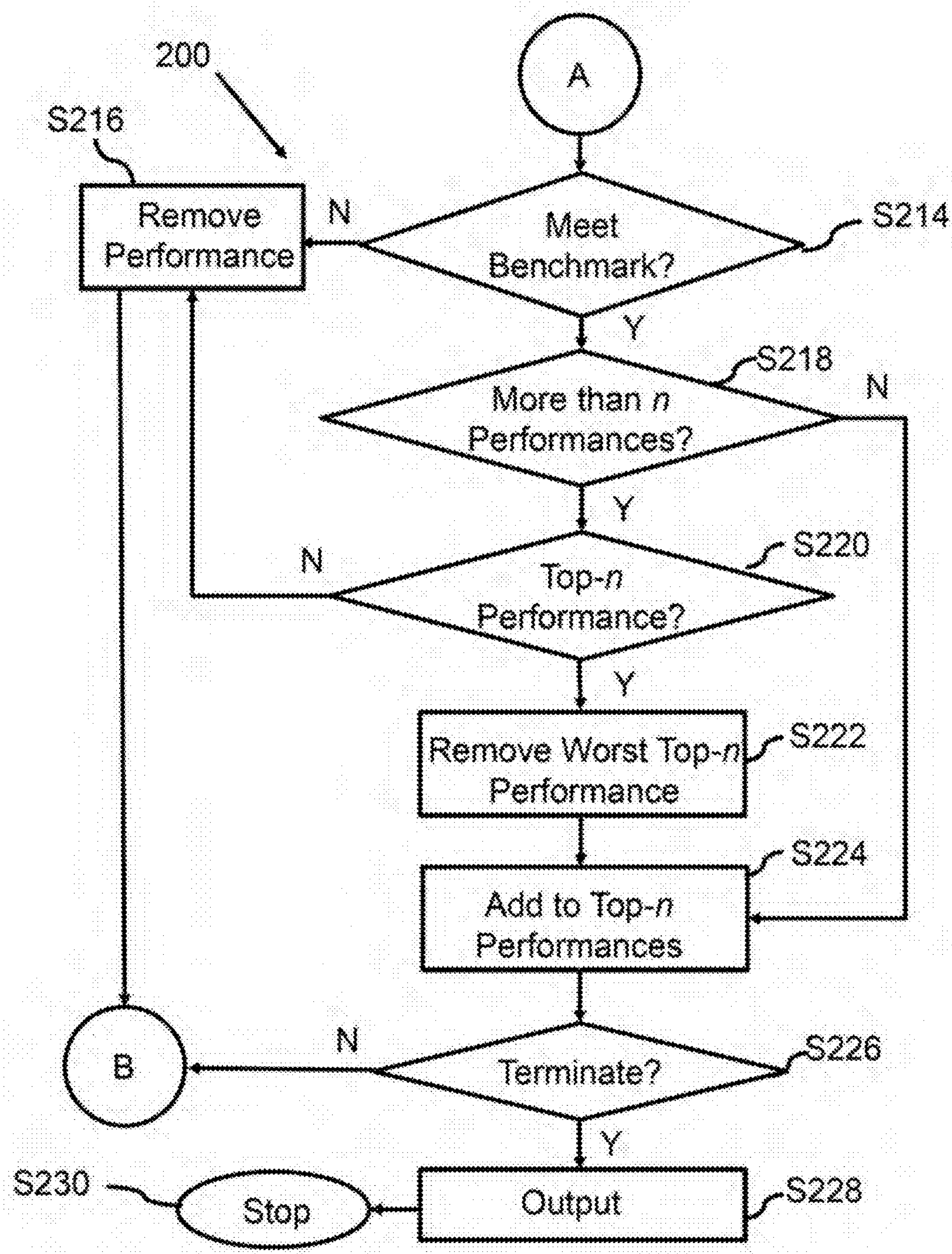

200
A
S216
Remove Performance
N
Meet Benchmark?
S214
Y
S218
More than n Performances?
N
Y
S220
N
Top-n Performance?
Y
Remove Worst Top-n Performance
S222
S224
Add to Top-n Performances
N
B
Terminate?
S226
Y
S230
Stop
Output
S228

FIG. 2B

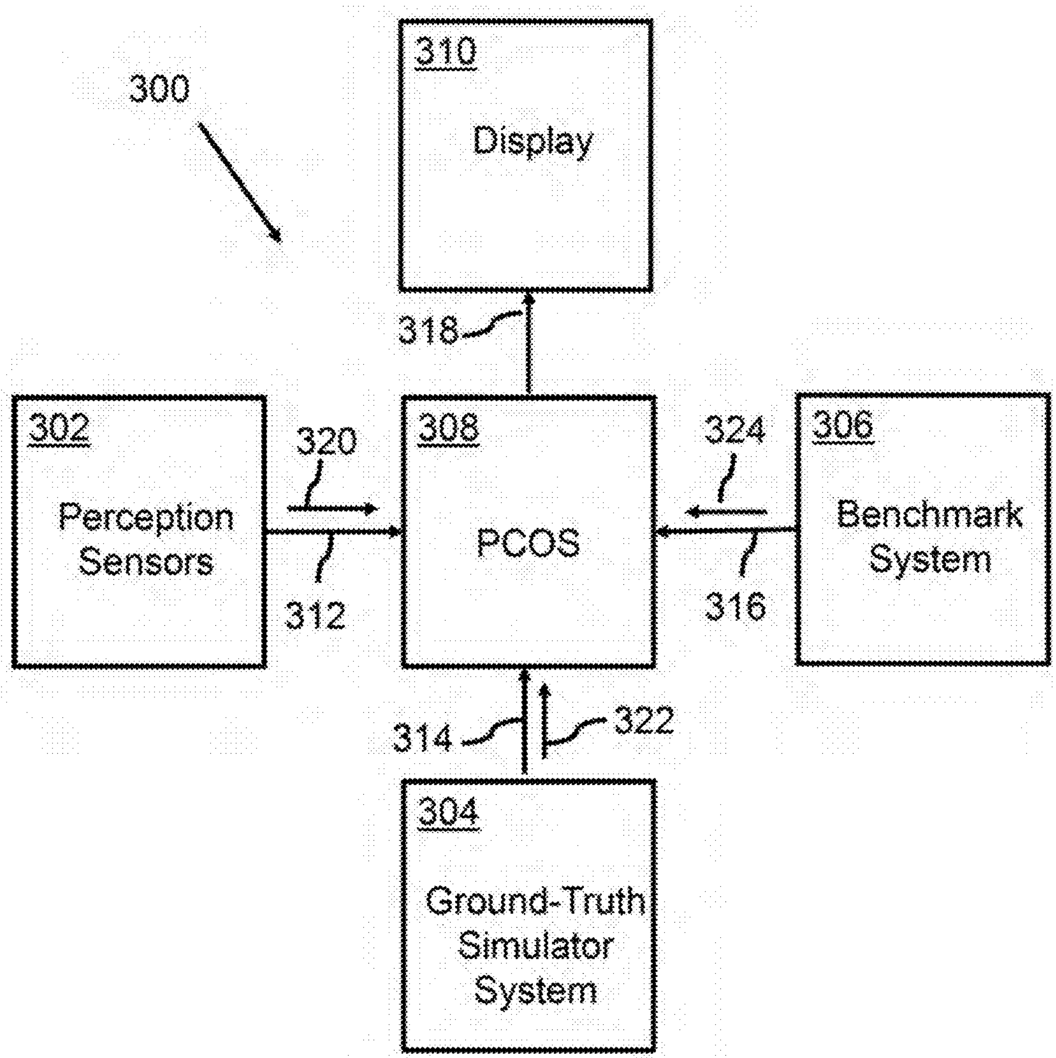
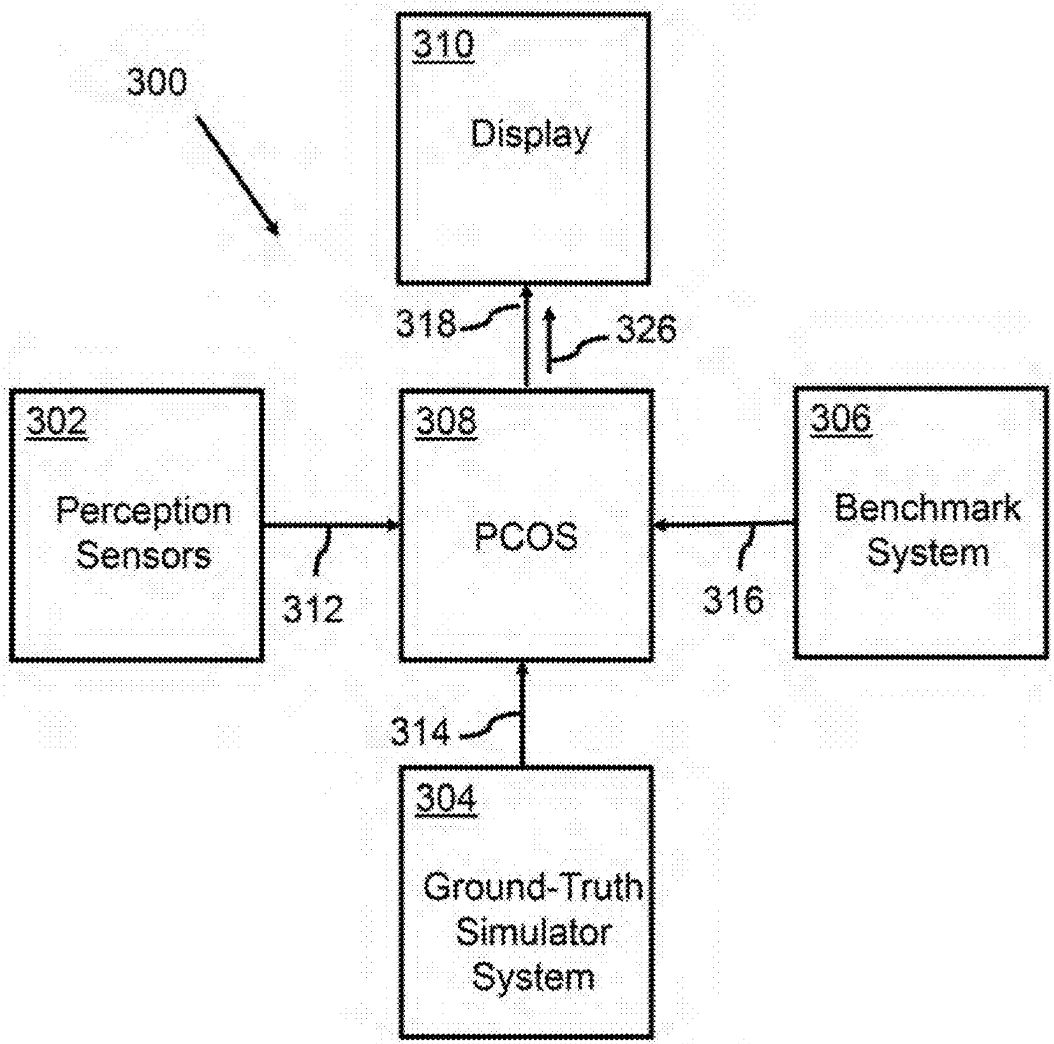

300
310
Display
318
302
Perception Sensors
320
312
308
PCOS
324
316
306
Benchmark System
314
322
304
Ground-Truth Simulator System

FIG. 3A



300
310
Display
318
326
302
Perception Sensors
312
308
PCOS
306
Benchmark System
316
314
304
Ground-Truth Simulator System


FIG. 3B

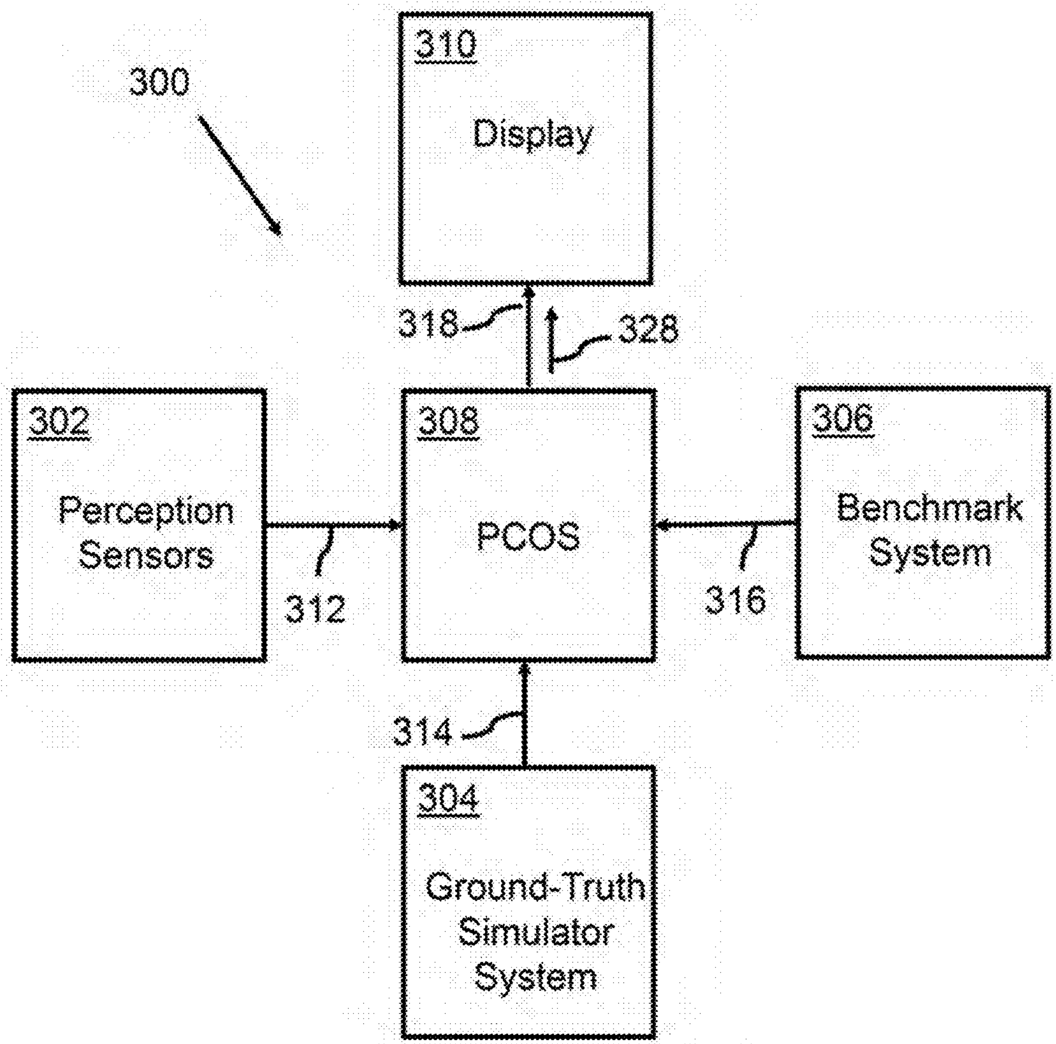
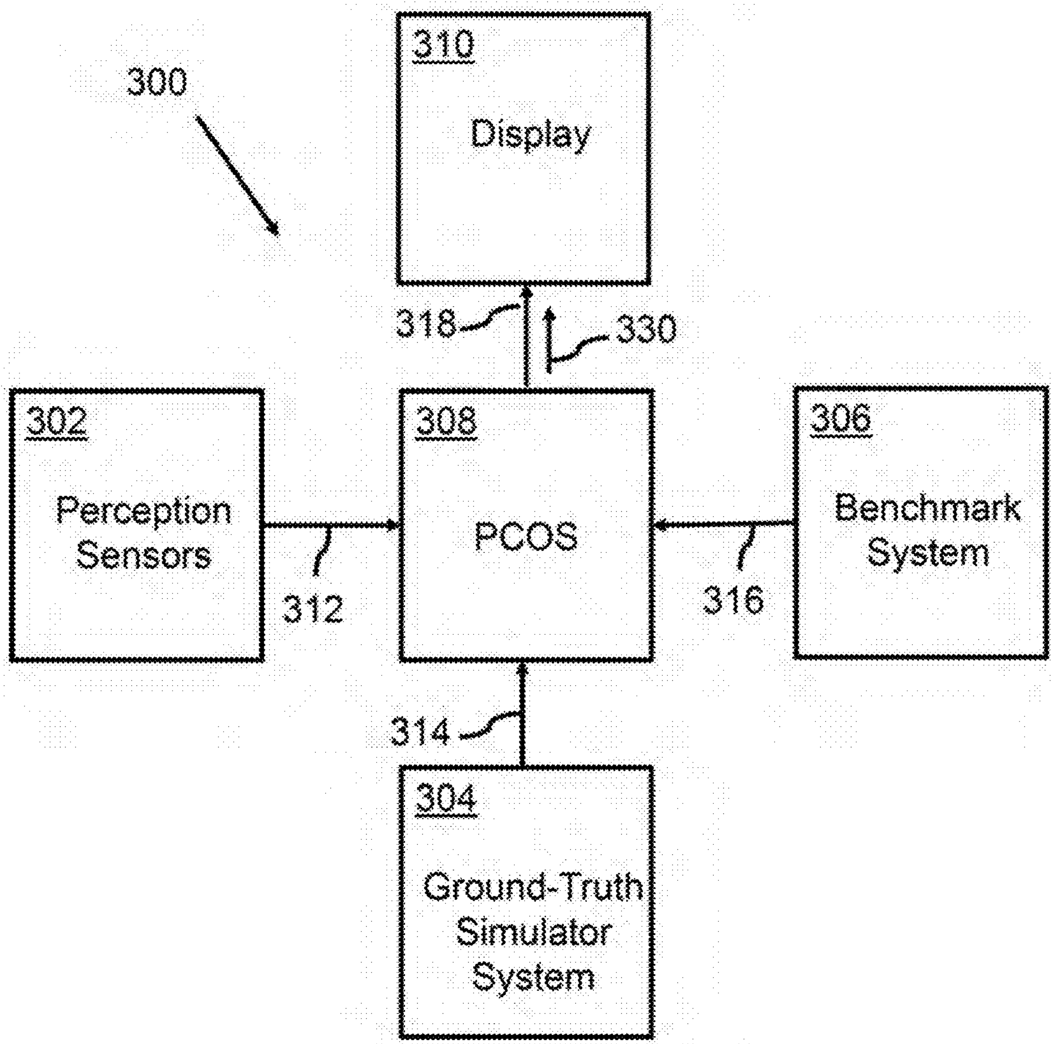
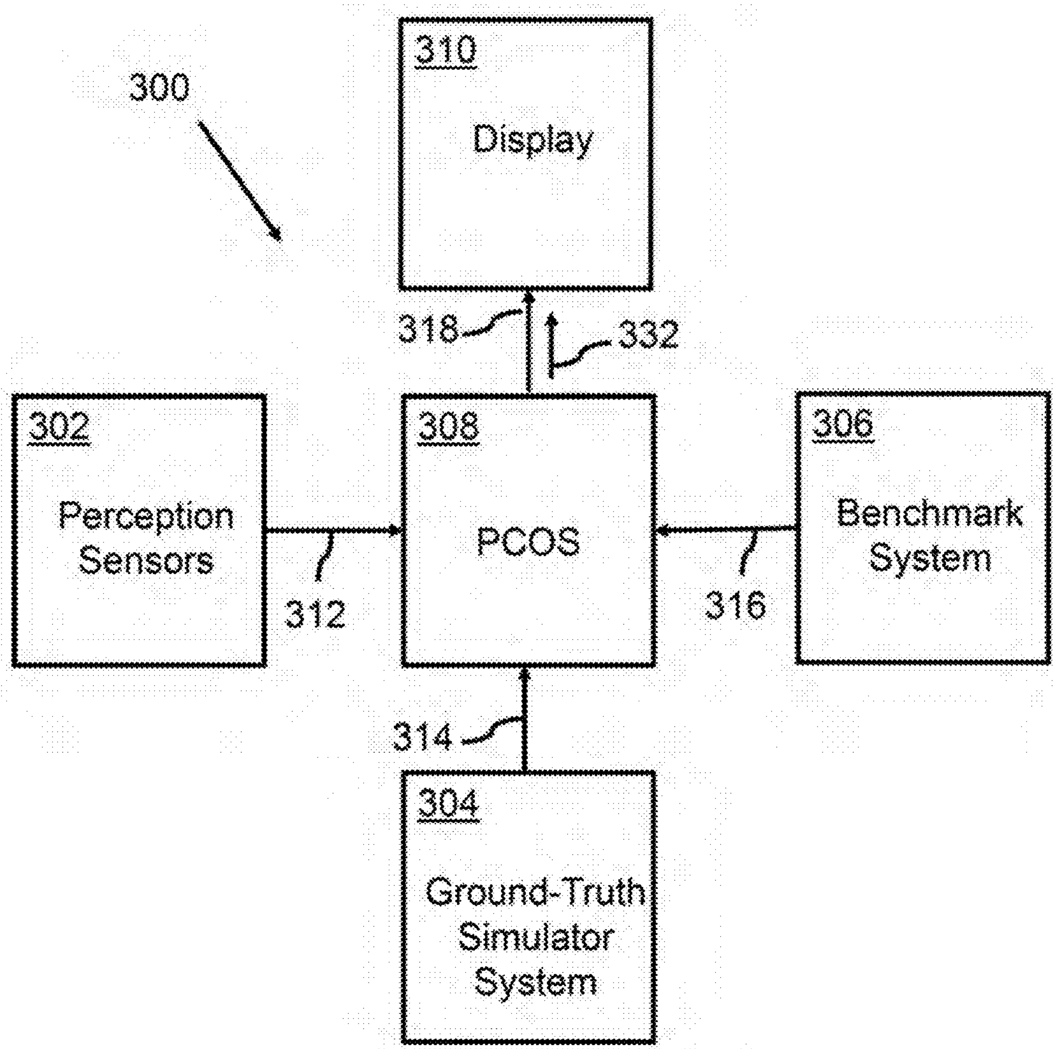

300
310
Display
318
328
302
Perception Sensors
308
PCOS
306
Benchmark System
312
316
314
304
Ground-Truth Simulator System

FIG. 3C



300
310
Display
318
330
302
Perception Sensors
312
308
PCOS
316
306
Benchmark System
314
304
Ground-Truth Simulator System


FIG. 3D



300
310
Display
318
332
302
Perception Sensors
312
308
PCOS
306
Benchmark System
316
314
304
Ground-Truth Simulator System


FIG. 3E

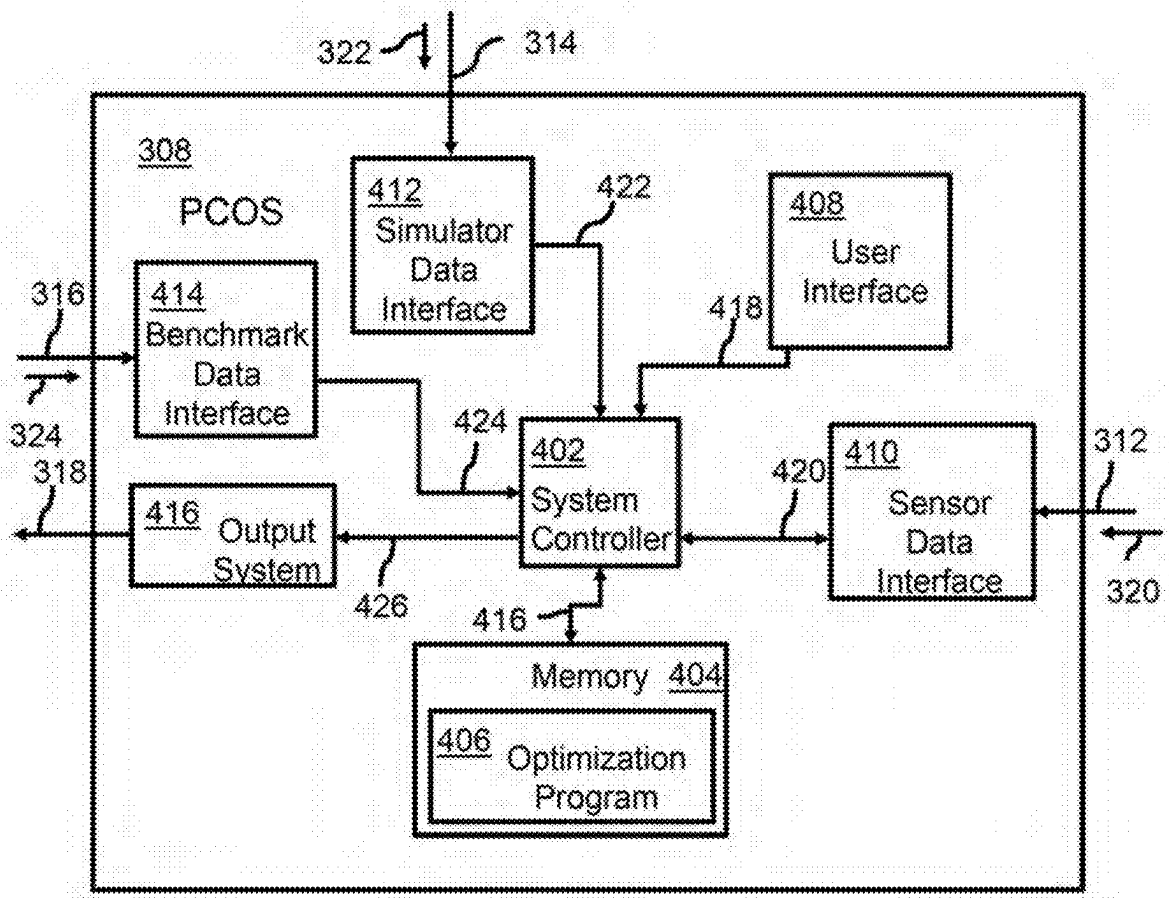

322
314
308
PCOS
412
Simulator
Data
Interface
422
408
User
Interface
316
414
Benchmark
Data
Interface
418
324
424
402
System
Controller
420
410
Sensor
Data
Interface
312
318
416
Output
System
426
416
320
Memory 404
406
Optimization
Program

FIG. 4A

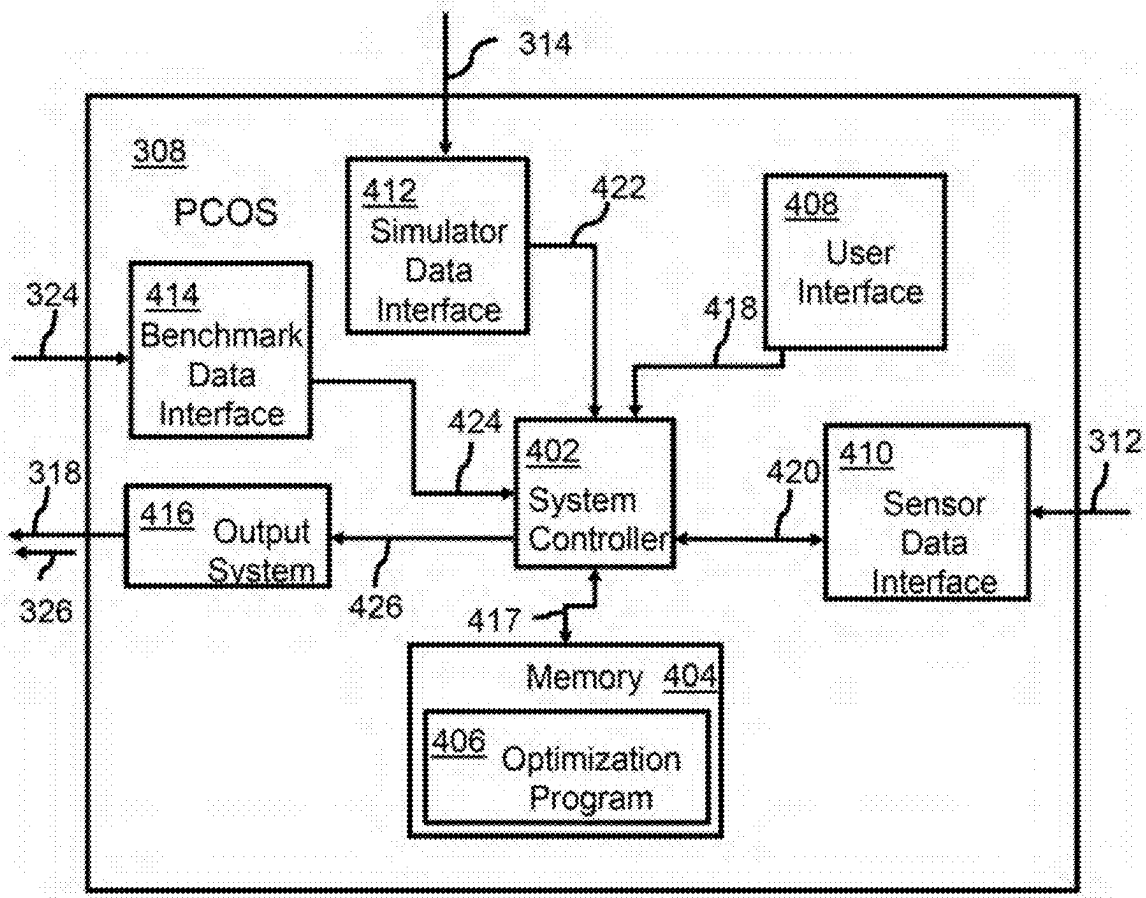

314
308
PCOS
412
Simulator
Data
Interface
422
408
User
Interface
324
414
Benchmark
Data
Interface
418
424
402
System
Controller
312
410
Sensor
Data
Interface
420
318
416
Output
System
326
426
417
Memory 404
406 Optimization
Program

FIG. 4B

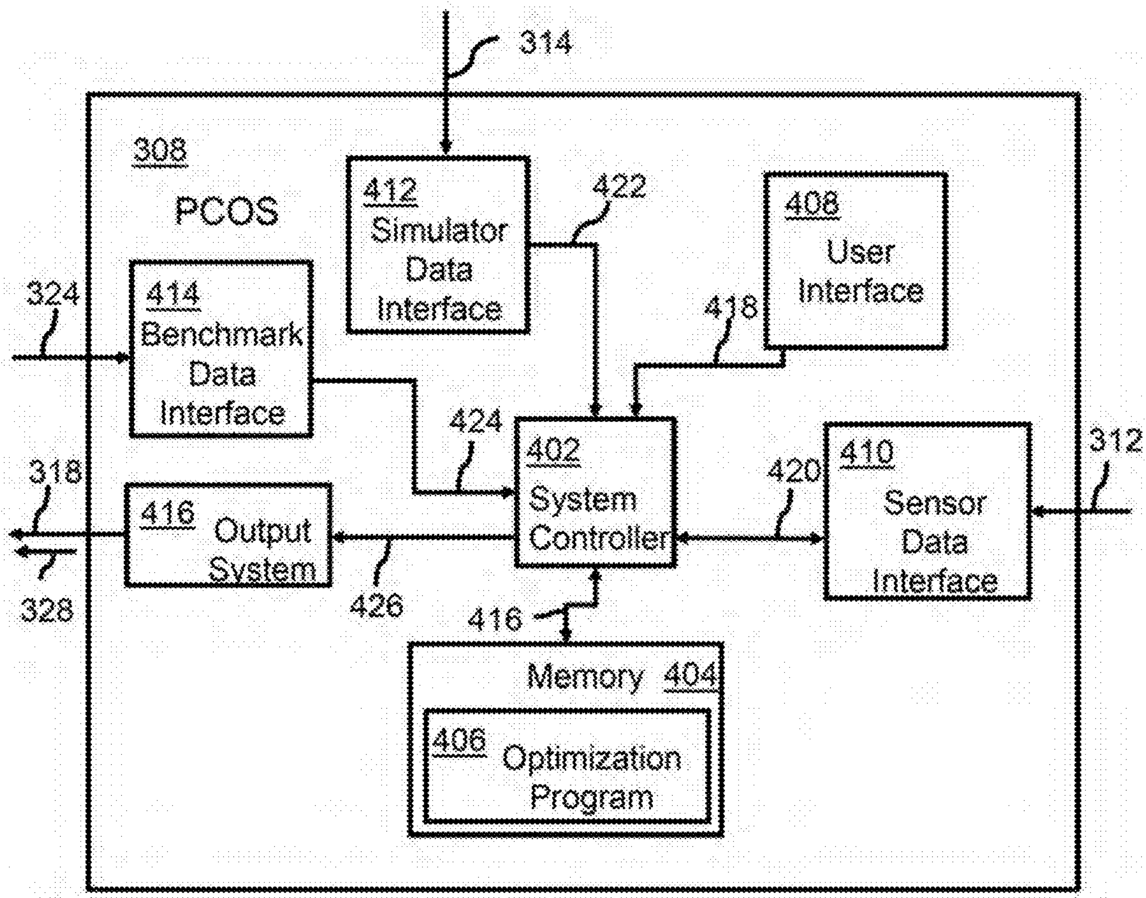

314
308
PCOS
412
Simulator
Data
Interface
422
408
User
Interface
324
414
Benchmark
Data
Interface
418
424
402
System
Controller
420
410
Sensor
Data
Interface
312
318
416
Output
System
328
426
416
Memory 404
406
Optimization
Program

FIG. 4C

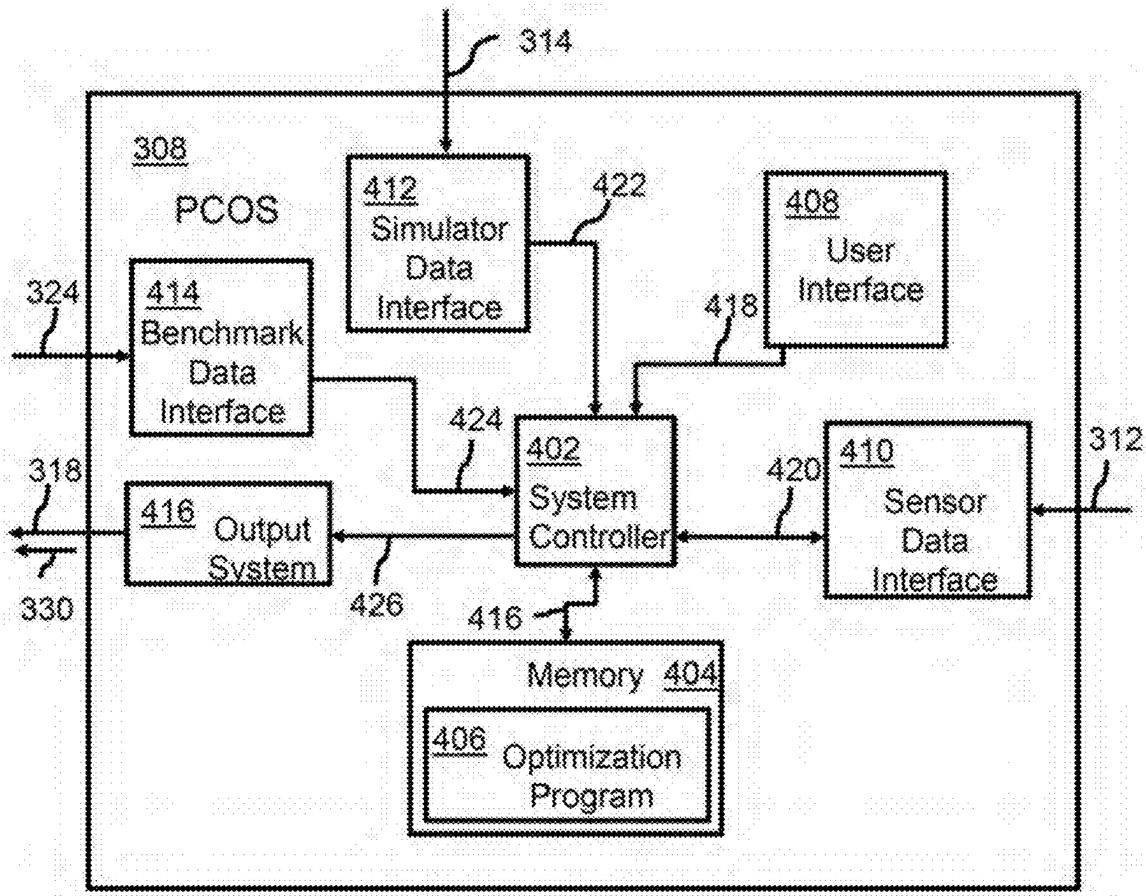

314
308
PCOS
412
Simulator
Data
Interface
422
408
User
Interface
324
414
Benchmark
Data
Interface
418
424
402
System
Controller
410
Sensor
Data
Interface
312
318
416
Output
System
420
426
330
416
Memory
404
406
Optimization
Program

FIG. 4D

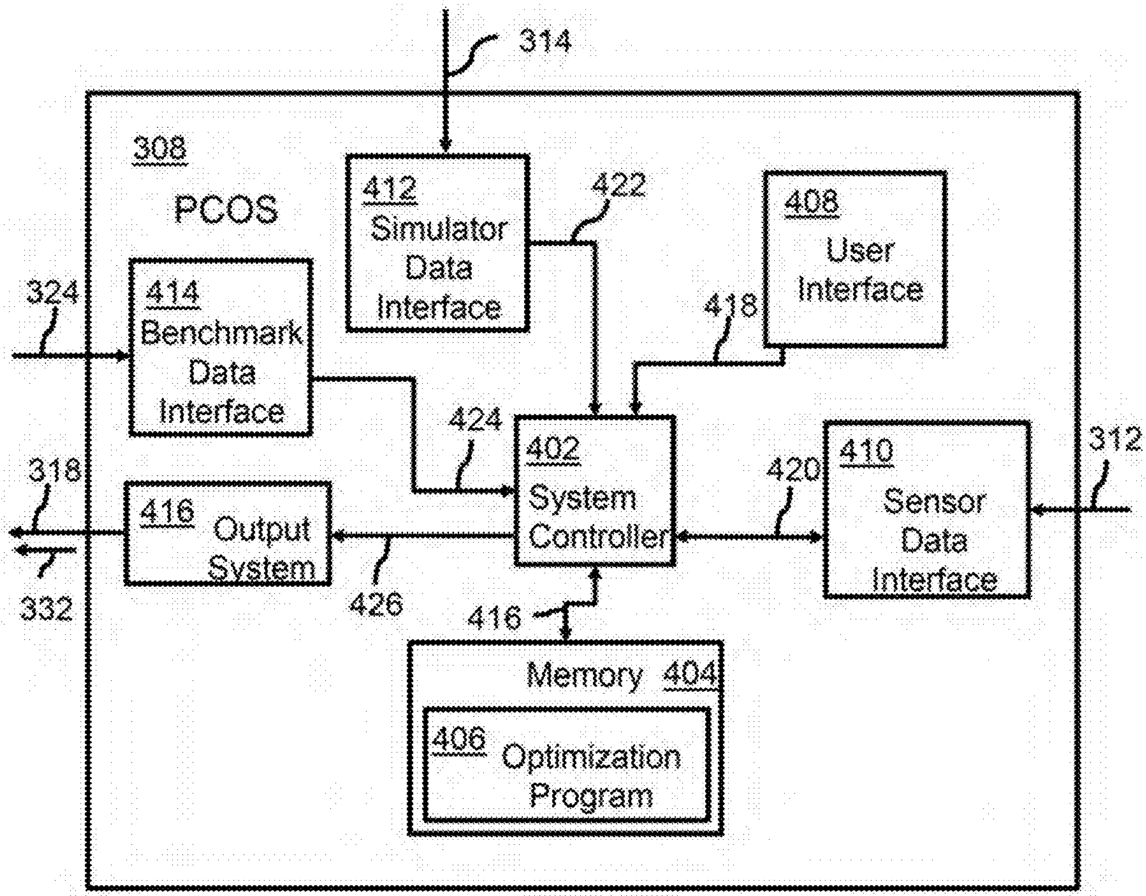

314
308
PCOS
412
Simulator
Data
Interface
422
408
User
Interface
324
414
Benchmark
Data
Interface
418
424
402
System
Controller
410
Sensor
Data
Interface
312
318
416
Output
System
420
332
426
416
Memory 404
406
Optimization
Program

FIG. 4E

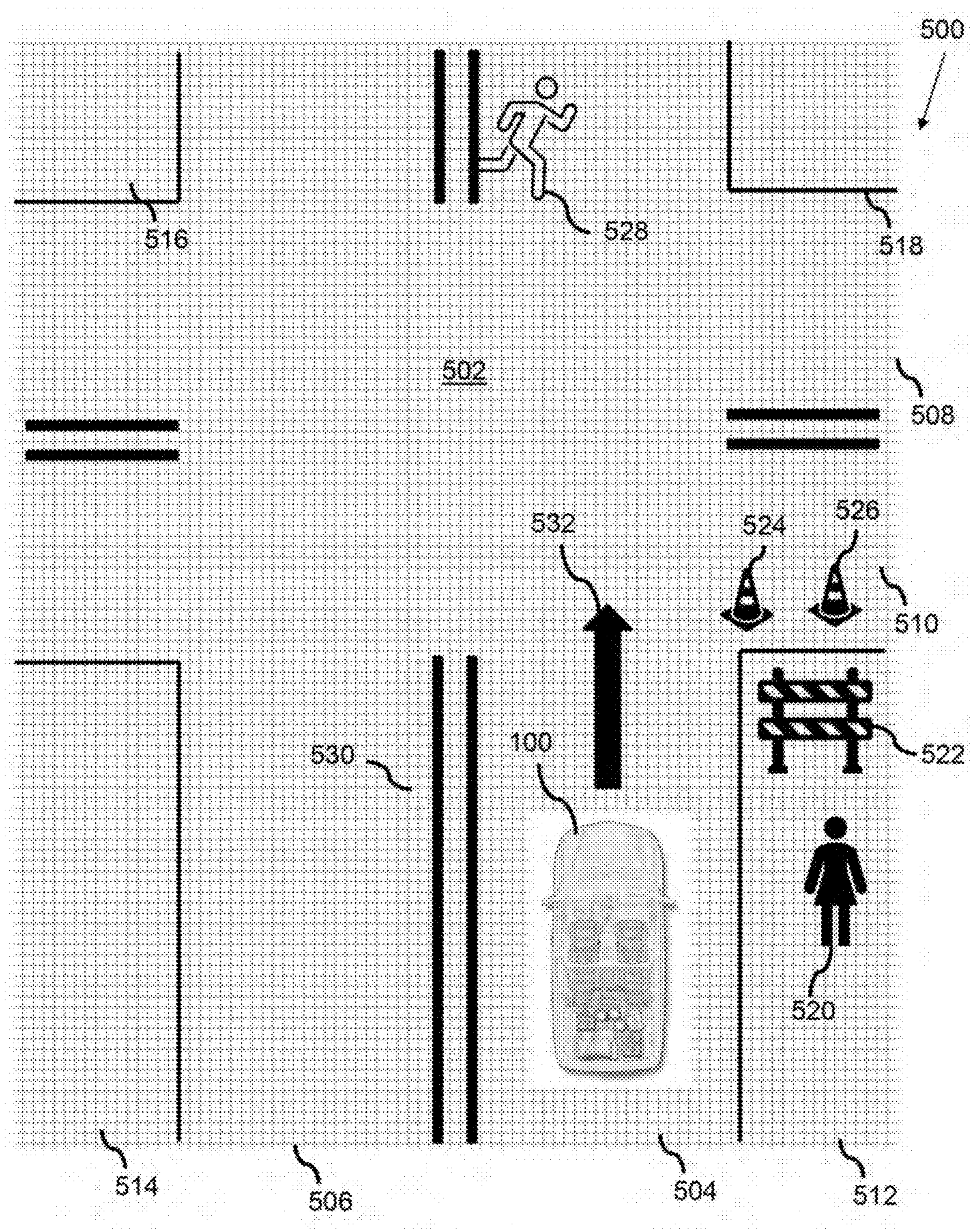

500
516
528
518
502
508
532
524
526
510
522
530
100
520
514
506
504
512

FIG. 5A

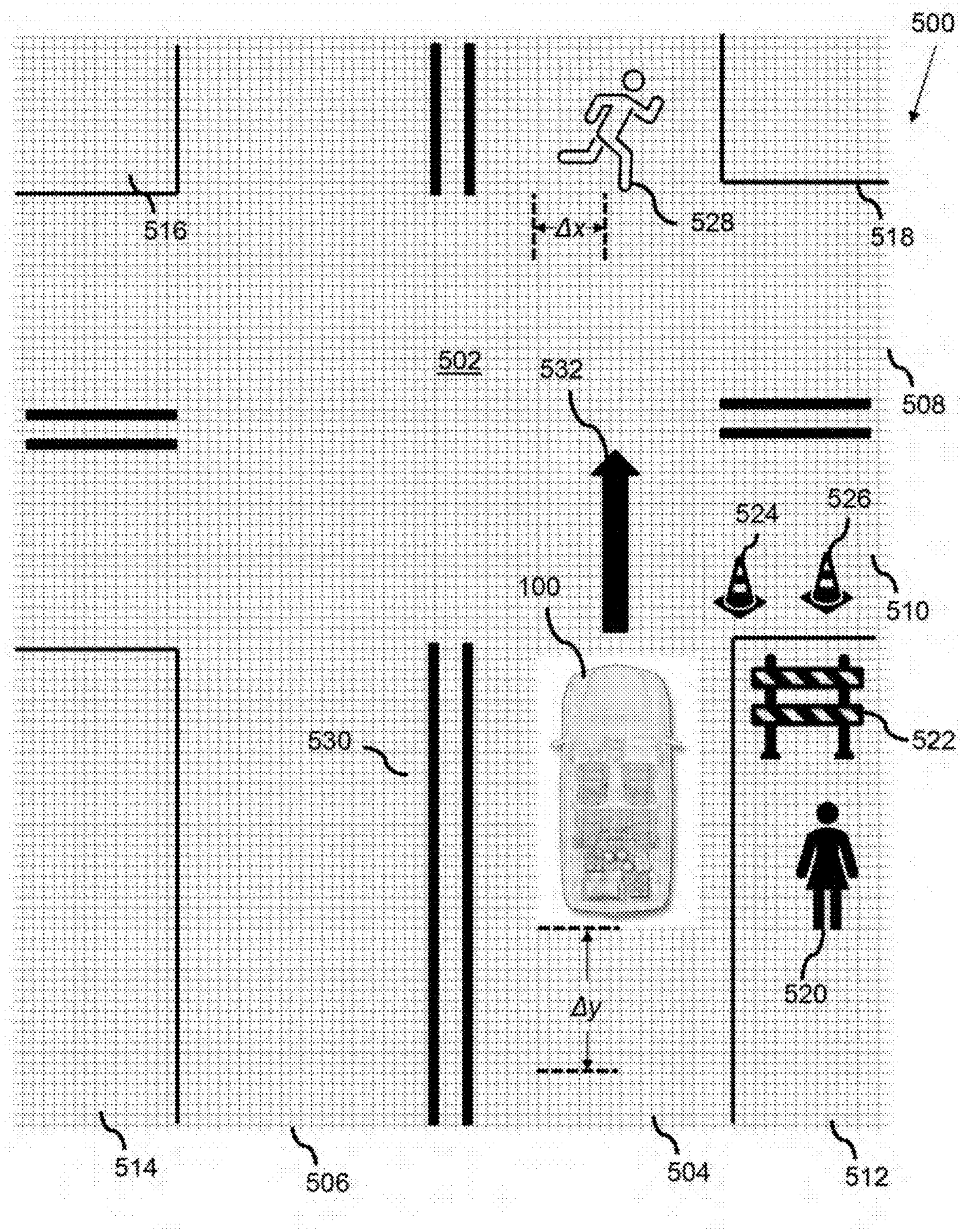

500
516
528
Δx
518
502
532
508
524
526
510
100
522
530
520
Δy
514
506
504
512

FIG. 5B

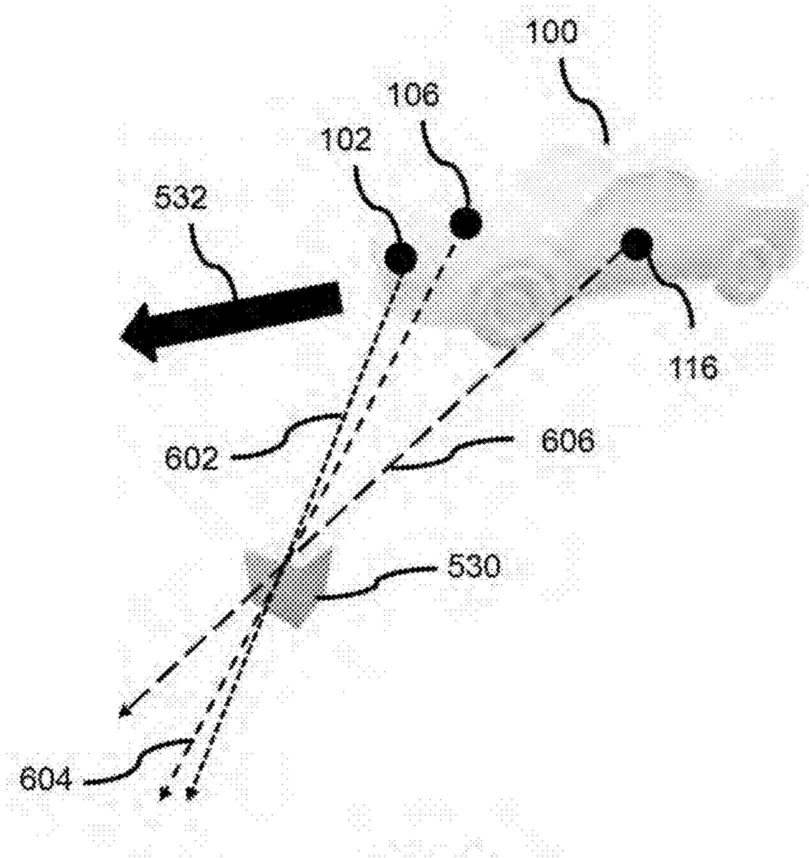



FIG. 6A

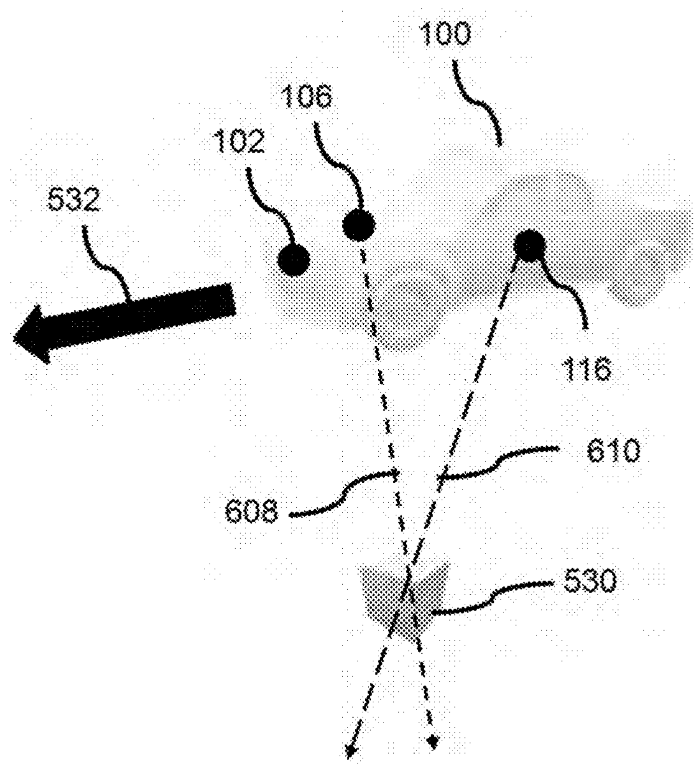



FIG. 6B

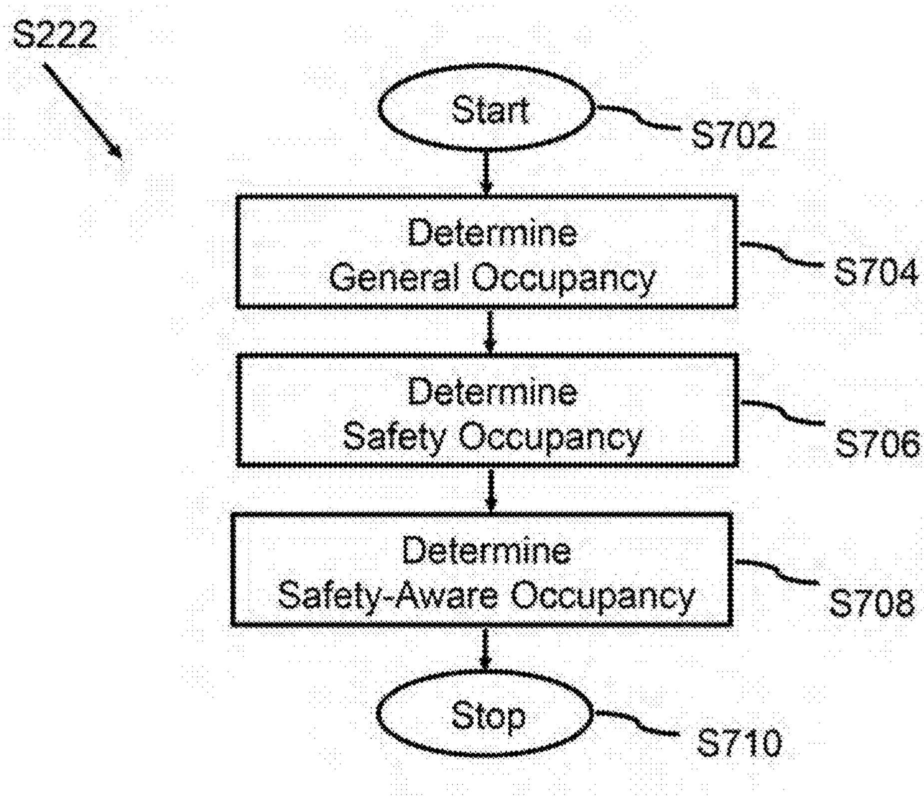

S222
Start
S702
Determine
General Occupancy
S704
Determine
Safety Occupancy
S706
Determine
Safety-Aware Occupancy
S708
Stop
S710

FIG. 7

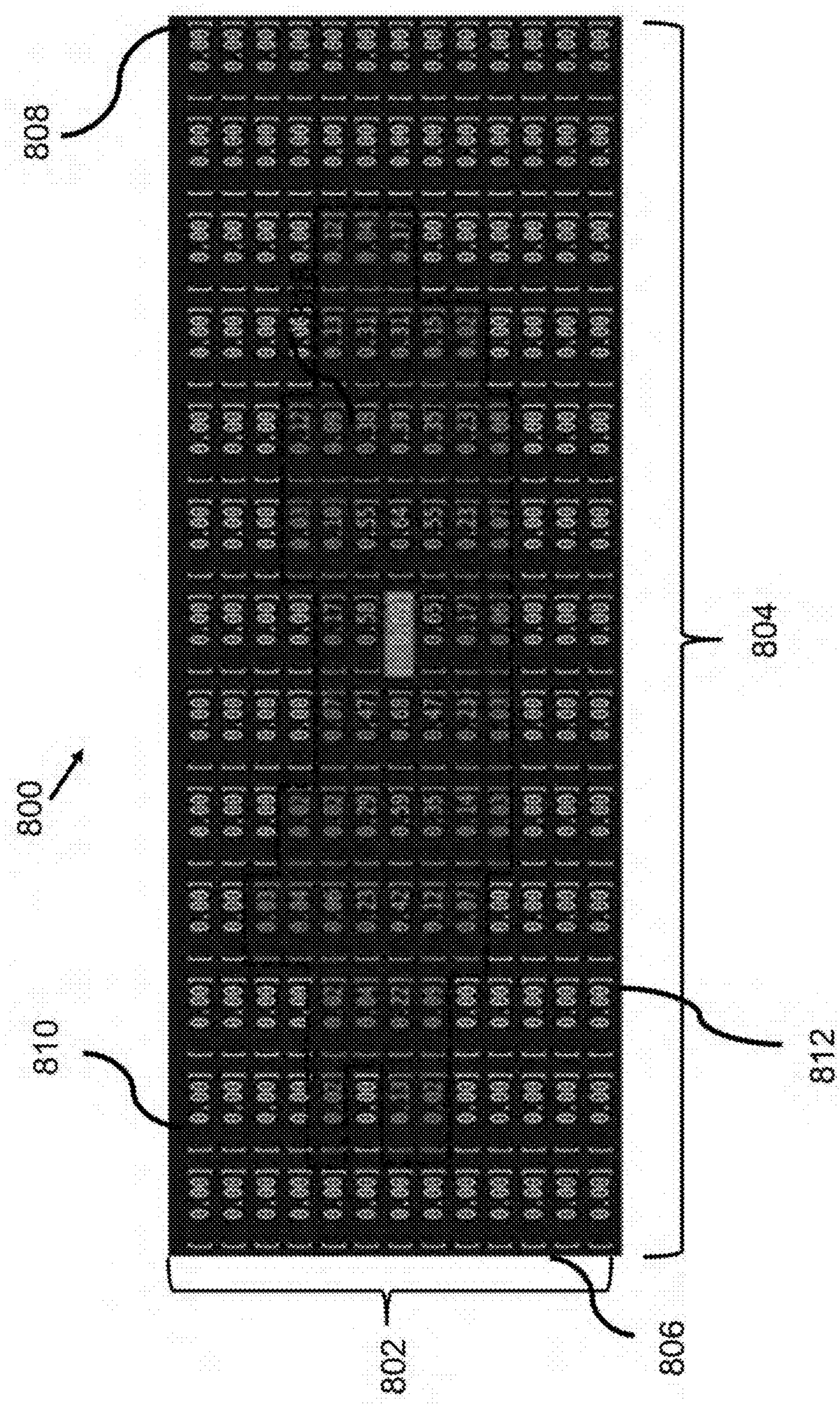



FIG. 8

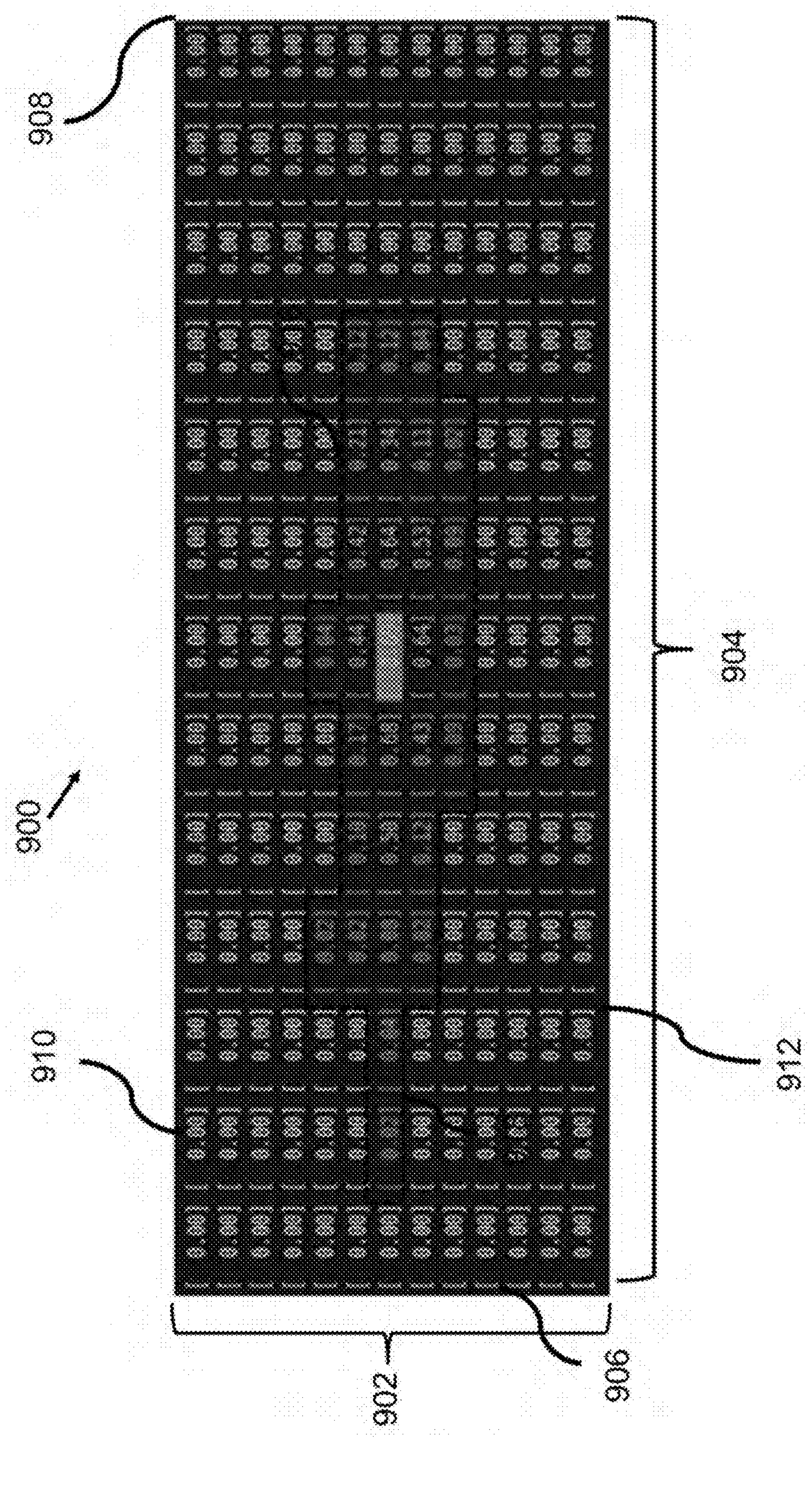



FIG. 9

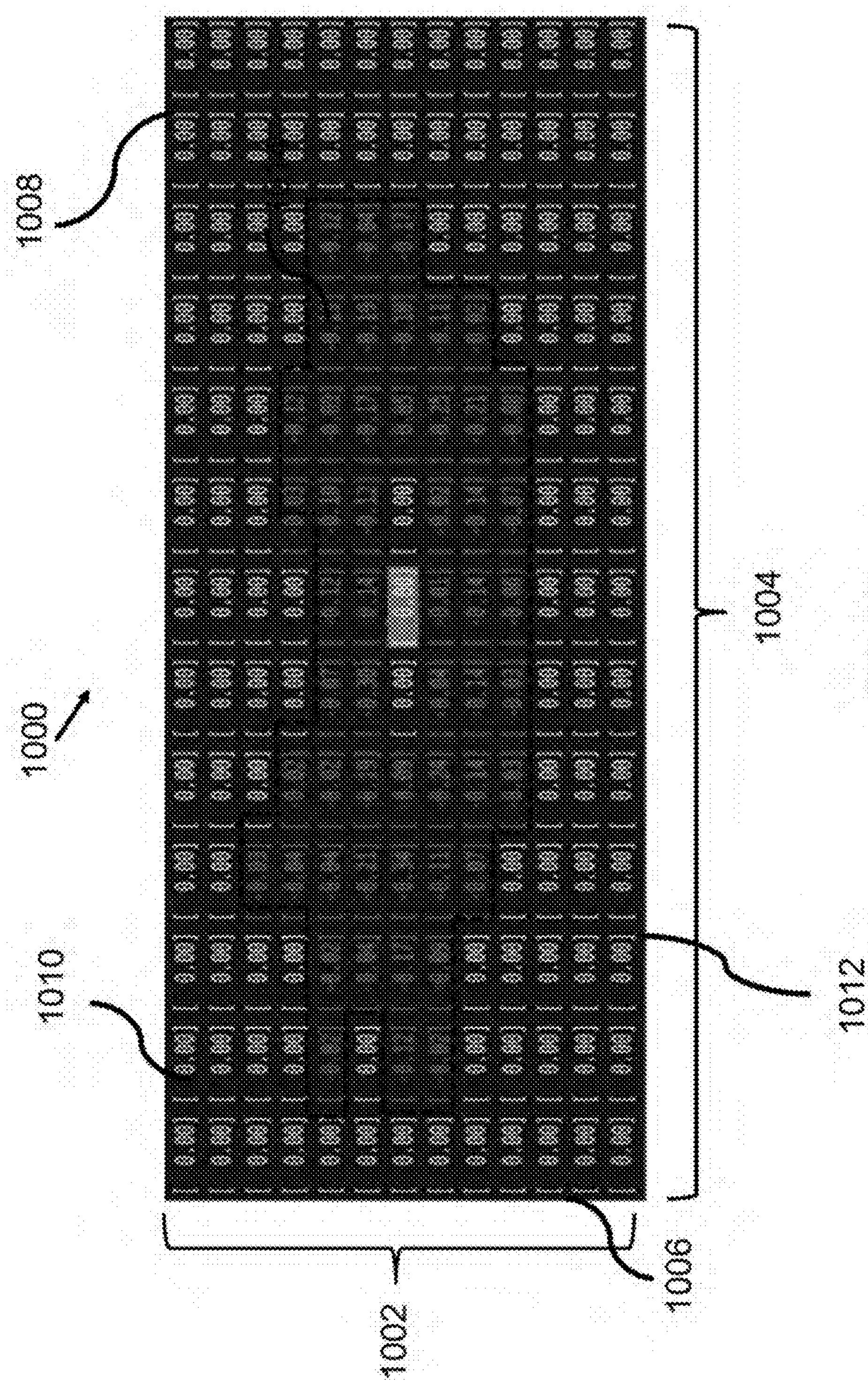



FIG. 10

# SYSTEM AND METHOD FOR OPTIMIZATION OF VEHICLE PERCEPTION SENSOR CONFIGURATION

## BACKGROUND

**[0001]** Autonomous vehicles (AVs) may require perception of the world in order to make intelligent control decisions. Generally, perception systems are composed of a heterogeneous set of sensors, including, e.g., cameras, ultrasonic/sonar, radar, and LiDAR.

## SUMMARY

**[0002]** An aspect of the present disclosure is drawn to a perception sensor configuration optimization system including: a memory having instructions stored therein; and a processor configured to execute the instructions stored in the memory to cause the perception sensor configuration optimization system to: receive perception sensor input data of an autonomous vehicle, wherein the perception sensor input data corresponds to respective measured location, orientation, and type of a plurality of perception sensors of the autonomous vehicle; receive ground-truth information input data of the autonomous vehicle, wherein the ground-truth information input data corresponds to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle; determine, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and output a safety-aware occupancy signal based on the safety occupancy.

**[0003]** In some embodiments of this aspect, the processor is further configured to execute the instructions stored in the memory to cause the perception sensor configuration optimization system to: determine, based on the perception sensor input data and the ground-truth information input data, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by: splitting a region around the autonomous vehicle into a plurality of voxels; and calculating a respective likelihood of the at least one obstacle occupying each voxel of the plurality of voxels; determine, based on the general occupancy of the at least one obstacle and the safety occupancy of the at least one obstacle, a safety-aware occupancy; and output the safety-aware occupancy signal based on the safety-aware occupancy. In some of these embodiments, the processor is further configured to execute the instructions stored in the memory to cause the perception sensor configuration optimization system to output, to a display, the safety-aware occupancy signal to display a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle. In some of these embodiments, the processor is further configured to execute the instructions stored in the memory to additionally cause the perception sensor configuration optimization system to compare the safety-aware occupancy with benchmark data corresponding to predetermined thresholds of acceptability.

**[0004]** In some embodiments of this aspect, the processor is further configured to execute the instructions stored in the memory to cause the perception sensor configuration optimization system to determine the safety occupancy by: splitting a region around the autonomous vehicle into a plurality of voxels; calculating, for each voxel of the plurality of voxels, a respective likelihood of an obstacle occupation; and establishing constraints to a presence of an obstacle.

**[0005]** In some embodiments of this aspect, the plurality of perception sensors of the autonomous vehicle include a distance perception sensor configured to provide distance information input data of the at least one obstacle, and the processor is further configured to execute the instructions stored in the memory to cause the perception sensor configuration optimization system to determine the safety occupancy by determining, based on the distance information input data and the ground-truth information input data, the safety occupancy of the at least one obstacle within the area around the autonomous vehicle.

**[0006]** In some embodiments of this aspect, the perception sensor configuration optimization system further includes a user interface configured to enable a user to modify the perception sensor input data.

**[0007]** Another aspect of the present disclosure is drawn to a method including: receiving perception sensor input data of an autonomous vehicle, the perception sensor input data corresponding to respective measured location, orientation, and type of each of a plurality of perception sensors of the autonomous vehicle; receiving ground-truth information input data of the autonomous vehicle, the ground-truth information input data corresponding to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle; determining, via a processor configured to execute instructions stored in a memory, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and outputting, via the processor, a safety-aware occupancy signal based on the safety occupancy.

**[0008]** In some embodiments of this aspect, the method further includes: determining, via the processor, based on the perception sensor input data and the ground-truth information input data, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by: splitting, via the processor, a region around the autonomous vehicle into a plurality of voxels; and calculating, via the processor, a respective likelihood of the at least one obstacle occupying each voxel of the plurality of voxels; determining, via the processor, based on the general occupancy of the at least one obstacle and the safety occupancy of the at least one obstacle, a safety-aware occupancy; and outputting, via the processor, the safety-aware occupancy signal based on the safety-aware occupancy. In some of these embodiments, the method further includes displaying, via a display, a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle. In some of these embodiments, the method further includes: receiving, via the processor and from a benchmark system, benchmark data corresponding to predetermined thresholds of acceptability; and comparing, via the processor, the safety-aware occupancy with the benchmark data.

**[0009]** In some embodiments of this aspect, determining the safety occupancy includes: splitting, via the processor, a region around the autonomous vehicle into a plurality of voxels; calculating, via the processor and for each voxel of the plurality of voxels, a respective likelihood of an obstacle occupation; and establishing, via the processor, constraints to a presence of any obstacle.

**[0010]** In some embodiments of this aspect, determining the safety occupancy of obstacles within the area around the autonomous vehicle includes determining, via the processor and based on distance information input data of the at least one obstacle and the ground-truth information input data, the safety occupancy of obstacles within the area around the autonomous vehicle.

**[0011]** In some embodiments of this aspect, the method further includes enabling, via a user interface, a user to modify the perception sensor input data.

**[0012]** Another aspect of the present disclosure is drawn to a non-transitory, computer-readable media having computer-readable instructions stored thereon, which, when executed across one or more processors, causes at least a portion of the one or more processors to perform operations including: receiving perception sensor input data of an autonomous vehicle, the perception sensor input data corresponding to respective measured location, orientation, and type of a plurality of perception sensors of the autonomous vehicle; receiving ground-truth information input data of the autonomous vehicle, the ground-truth information input data corresponding to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle; determining, via a processor configured to execute instructions stored in a memory, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and outputting, via the processor, a safety-aware occupancy signal based on the safety occupancy.

**[0013]** In some embodiments of this aspect, the operations further include: determining, based on the perception sensor input data and the ground-truth information input data, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by: splitting a region around the autonomous vehicle into a plurality of voxels; and calculating a respective likelihood of the at least one obstacle occupying each voxel of the plurality of voxels; determining based on the general occupancy of the at least one obstacle and the safety occupancy of the at least one obstacle, a safety-aware occupancy; and outputting the safety-aware occupancy signal based on the safety-aware occupancy. In some of these embodiments, the operations further include displaying, via a display, a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle.

**[0014]** In some embodiments of this aspect, determining the safety occupancy includes: splitting a region around the autonomous vehicle into a plurality of voxels; calculating, for each voxel of the plurality of voxels, a respective likelihood of an obstacle occupation; and establishing constraints to a presence of an obstacle.

**[0015]** In some embodiments of this aspect, determining the safety occupancy of obstacles within the area around the autonomous vehicle includes determining, via distance information input data of the at least one obstacle and the ground-truth information input data, the safety occupancy of obstacles within the area around the autonomous vehicle.

**[0016]** In some embodiments of this aspect, the operations further include enabling, via a user interface, a user to modify the perception sensor input data.

**[0017]** The details of one or more example implementations are set forth in the accompanying drawings and the description below. Other possible example features and/or possible example advantages will become apparent from the description, the drawings, and the claims. Some implementations may not have those possible example features and/or possible example advantages, and such possible example features and/or possible example advantages may not necessarily be required of some implementations

## BRIEF SUMMARY OF THE DRAWINGS

**[0018]** The accompanying drawings, which are incorporated in and form a part of the specification, illustrate and explain examples. In the drawings:

**[0019]** FIG. **1**A illustrates an autonomous vehicle with a sensor configuration thereon;

**[0020]** FIG. **1**B illustrates the autonomous vehicle with a second sensor configuration thereon;

**[0021]** FIG. **2**A-B illustrates an example method for optimization of a perception sensor configuration in accordance with aspects of the present disclosure, wherein FIG. **2**A illustrates a first portion of the example method and FIG. **2**B illustrates a second portion of the example method;

**[0022]** FIG. **3**A illustrates an example perception sensor configuration optimization system in accordance with aspects of the present disclosure at a time $t_3$;

**[0023]** FIG. **3**B illustrates the example perception sensor configuration optimization system of FIG. **3**A at a time $t_4$;

**[0024]** FIG. **3**C illustrates the example perception sensor configuration optimization system of FIG. **3**A at a time $t_5$;

**[0025]** FIG. **3**D illustrates the example perception sensor configuration optimization system of FIG. **3**A at a time $t_6$;

**[0026]** FIG. **3**E illustrates perception sensor configuration optimization system of FIG. **3**A at a time $t_7$;

**[0027]** FIG. **4**A illustrates an example of perception sensor configuration optimization system (PCOS) in accordance with aspects of the present disclosure at a time $t_3$;

**[0028]** FIG. **4**B illustrates the example PCOS of FIG. **4**B;

**[0029]** FIG. **4**C illustrates the example PCOS of FIG. **4**C;

**[0030]** FIG. **4**D illustrates the example PCOS of FIG. **4**D;

**[0031]** FIG. **4**E illustrates the example PCOS of FIG. **4**A at a time

**[0032]** FIG. **5**A illustrates a plan view of an example autonomous vehicle driving through a portion of a city at a time $t_0$;

**[0033]** FIG. **5**B illustrates a plan view of the autonomous vehicle driving through the portion of the city of FIG. **5**A at a time $t_1$.

**[0034]** FIG. **6**A illustrates a perspective view of an autonomous vehicle and a scanning of a voxel of FIG. **5**A;

**[0035]** FIG. **6**B illustrates a perspective view of the autonomous vehicle and a scanning of a voxel of FIG. **5**B;

**[0036]** FIG. **7** illustrates a more detailed view of the process of finding safety information-gain;

**[0037]** FIG. **8** illustrates an example safety occupancy grid in accordance with aspects of the present disclosure;

**[0038]** FIG. **9** illustrates an example general occupancy grid in accordance with aspects of the present disclosure; and

[0039] FIG. 10 illustrates an example safety-aware occupancy grid in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0040] The perception system that an AV employs is currently configured through trial-and-error; in other words, there is no process for optimizing their configuration given some constraints, e.g., safety/risk assessment capabilities.

[0041] Some prior technologies have investigated the optimal LiDAR placement for general perception metrics. Further, different conventional general perception configurations have been reviewed; however, such reviews do not investigate whether these configurations are optimal with respect to a given set of criteria.

[0042] Furthermore, some prior systems focus on safety-aware autonomous systems for the purpose of detecting pedestrians. These prior art systems use time-to-collision as a metric for assessing a deep neural network's performance for detecting critical safety scenarios. Finally, a branch of research investigates the assume-guarantee relationships for autonomous system design. This line of research seeks to understand the guarantees a system may provide given assumptions of a scenario's composition. Implicitly, this technology supports a broad range of safety applications through formal verification.

[0043] It may be beneficial to have a system and method for optimizing a perception system configuration of an AV given some constraints, e.g., safety/risk assessment capabilities

[0044] A system, computer-readable media, and method in accordance with aspects of the present disclosure optimizes a perception system configuration of an AV given some constraints, e.g., safety/risk assessment capabilities.

[0045] An example and non-limiting purpose of a system, computer-readable media, and method in accordance with aspects of the present disclosure is to determine (given some risk assessment constraint) the optimal configuration of the sensors of a perception system for providing risk assessment capabilities. A system, computer-readable media, and method in accordance with aspects of the present disclosure was developed with the goal of optimizing the perception of pedestrians and ensuring their safety, but the general procedure is applicable to any autonomous or automated system, such as ground or aerial mobile robots, infrastructure monitoring, etc.

[0046] Compared to the systems discussed above, the example and non-limiting problem to be solved is to investigate the optimal perception configuration (i.e., general to all perception sensors) for safety perception metrics. In other words, in accordance with an example and non-limiting aspect of the present disclosure, an optimal perception configuration may be found consisting of all sensors and consisting of configurations that might not be deployed in the world today. A framework in accordance with aspects of the present disclosure also builds on other AV metrics that incorporate safety context of the vehicle. For instance, some conventional research investigates task-aware risk estimation for the purpose of path planning. The objective of this research is to mitigate the impact of perception failures, which is orthogonal to the objective of a system and method in accordance with aspects of the present disclosure. Additionally, this research is not presently compatible with the goal of finding an optimal perception configuration.

[0047] Unlike the technologies discussed above, a system and method in accordance with aspects of the present disclosure seeks to do at least two example and non-limiting things differently.

[0048] First, in accordance with aspects of the present disclosure, a more general optimization function is used that considers perception configurations that are not on the road today, making a system and method in accordance with aspects of the present disclosure capable of finding optimal sensor configurations that will improve the performance of AVs.

[0049] Second, in accordance with aspects of the present disclosure, the scope of a proxy metric is reduced by introducing a maximum information gain of information related to safety-performance.

[0050] While other systems also seek to use metrics as a way of assessing safety performance, they do not make use of a proxy metric. The main example and non-limiting advantage of a proxy metric is that it may not require the training of a large machine learning model, and subsequently may not require the computation of various perception (computer vision) model metrics. This is advantageous because, at least in part, doing so is computationally expensive even for just one perception configuration. Being able to iterate over an arbitrary number of configurations and computing the safety performance of each is a tremendous improvement.

[0051] As will be described in greater detail below, aspect of the present disclosure presents a high-level algorithm for detecting safety-critical events and then a more specific algorithm that makes use of conditions (i.e., specifications) for which the perception system should optimize.

[0052] In an example operation of a system and method in accordance with aspects of the present disclosure includes, first, providing to an optimization function, a set of candidate perception sensors, where they can be located, and how they can be oriented.

[0053] In one example embodiment, an optimization function that is solved in accordance with aspects of the present disclosure is shown below; it is formulated as a Binary Integer Program (BIP). A BIP, also known as a zero-one linear program, is a type of mathematical optimization problem where the variables are restricted to be either 0 or 1. This means that the decision variables can only take on two possible values, representing a "yes" or "no" decision. BIPs are a subset of integer programming problems, which are optimization problems where the variables are restricted to being integers. It should be noted that any known optimization procedure may be used that optimizes where a set of candidate perception sensors, where they can be located, and how they can be oriented.

[0054] In any event, with the BIP, the information gain of the region around the vehicle, $H(V)$, is maximized and is conditioned on some sensor configuration $x$. The BIP is constrained by all $N$ sensors being used only once and all $M$ locations sensors can be located within their set capacity $c_j$. The BIP also considers $P$ different orientations that a sensor can be in at each location.

$$\max_x S_{AMIG} = -H(V \mid x) \tag{1}$$

$$\text{s.t. } \sum_{j=1}^{M} \sum_{k=1}^{P} x_{i,j,k} \le 1 \;\; \forall\, i = 1, \ldots, N \tag{2}$$

$$\sum_{i=1}^{N} \sum_{k=1}^{P} x_{i,j,k} \le c_j \;\; \forall\, j = 1, \ldots, M \tag{3}$$

$$\text{wherein } x_{i,j,k} \in \{0,1\} \;\; \forall\, i = 1, \ldots, Nj = 1, \ldots, Mk = 1, \ldots, P$$

**[0055]** Equation 1 is the optimization function. A goal is to maximize the safety-aware information-gain. Equation 2 is the first constraint. Each sensor may only be used at most once (i.e., it may be used or not used). Equation 3 is the second constraint, wherein the number of sensors that occupy a specific location (from j=1, . . . , M) does not exceed the capacity $c_j$.

**[0056]** The scenarios under consideration are simulated and distance information input data of the at least one obstacle from an ego vehicle, the location and distance of other vehicles with respect to the ego vehicle, and information related to the weather is collected as a priori ground truth information input data. The ground truth information input data corresponds to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle. It should be noted that other data may be used in accordance with aspects of the present disclosure. In some embodiments, the distance information input data of the at least one obstacle includes at least one of LiDAR, radar, sonar, and camera data.

**[0057]** A simulator output that provides the necessary perception and environment ground truth information input data may be used for verifying the output of a system and method in accordance with aspects of the present disclosure. A non-limiting example of such a simulator includes CARLA, which is an open-source autonomous driving simulator.

**[0058]** Once the ground truth information input data is collected, it can be used to calculate a ground truth safety-related information gain score. Such ground truth information is easily accessible in simulators. It is also possible to engineer scenarios in real-world instrumented driving infrastructure such as Mcity, where there is user control over the data collection process. Mcity is a facility at the University of Michigan and that is purpose-built for testing connected and automated vehicles and technologies under controlled, realistic conditions.

**[0059]** The goal of the optimization function is to find a perception sensor configuration that maximizes the information gain, which is related to the difference between the entropy (uncertainty) of the ground truth perception observations, H(V), and the entropy (uncertainty) generated by a particular configuration H(V|x). H may be calculated by any known method, a non-limiting example of which is disclosed in Hu et al., "Investigating the Impact of Multi-LiDAR Placement on Object Detection for Autonomous Driving," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 2550-2559, the entire disclosure of which is incorporated herein by reference.

**[0060]** H(V) and $\hat{H}(v_i)$: are defined as:

$$H(V) := \sum_{v_i \in V} \hat{H}(v_i) \tag{4}$$

$$\hat{H}(v_i) = -\hat{p}(v_i)\log\hat{p}(v_i) - (1 - \hat{p}(v_i))\log(1 - \hat{p}(v_i)) \tag{5}$$

**[0061]** where $v_i$ is a voxel i. and $\hat{H}(v_i)$ is the information gain of a particular voxel ($v_i$) surrounding the vehicle. The expansion follows information entropy theory, from which the term information-gain is derived. The $\hat{p}(v_i)$, described below with reference to equation 6, is the ideal probability that an object will be detected over n frames. For example, if in the ideal world, an object would be detected in 1 out of 5 frames, then there is no expectation to perform better than that in with the real sensors.

**[0062]** A system and method in accordance with aspects of the present disclosure differs from prior art systems in the manner with which $\hat{p}(v_i)$ is calculated, which herein is defined as the probability that a given location near the vehicle will have a perception event occur that is of importance to safety. In a system and method in accordance with aspects of the present disclosure, $\hat{p}(v_i)$ is calculated using the ground-truth as well as the perception data that is collected for a particular sensor configuration. More specifically, a feature of a system and method in accordance with aspects of the present disclosure, as discussed in greater detail below, includes both how to calculate p ($v_i$) using the ground truth and how to calculate it given some sensor configuration x. During an optimization function process loop, the top-n configurations that are closest to the perfect safety metric score are extracted and further evaluated on more computationally expensive test scenarios. This "top-n" is, left to configured by the system designer to their evaluation needs. During this last step, the goal is to determine which candidate configuration performs the best; the proxy metric is an approximation, so this step allows the designer to verify the candidate configurations using the more robust and computationally expensive methods.

**[0063]** In some embodiments, an optimization loop may terminate based on a predetermined heuristic. In some of these embodiments, the optimization problem itself is a BIP, which is known to be computationally hard to solve. In accordance with aspects of the present disclosure, the predetermined heuristics enable iterating execution and determining when to terminate.

**[0064]** The purpose of a system and method in accordance with aspects of the present disclosure is to provide a framework for finding an optimal perception configuration on an autonomous vehicle for a general purpose (e.g., automated driving, pedestrian safety, vehicle crash prevention). There are two main features in accordance with aspects of the present disclosure: (1) creating a probabilistic occupancy grid based on whether particular perception information meets a defined set of safety specifications; (2) a formalized definition of an optimization problem for perception configuration based on a safety-aware information-gain that is constructed from the probabilistic occupancy grid.

**[0065]** As will be described in greater detail below, the safety-aware information gain for one iteration of the optimization loop is calculated. With one iteration completed, the process can generalize to an arbitrary number of iterations (until the set termination condition). For the non-limiting example proof-of-concept, two safety metrics were chosen to compare: (1) a general occupancy of pedestrians; and (2) a safety occupancy of pedestrians.

**[0066]** In accordance with aspects of the present disclosure, a general occupancy of pedestrians, is calculated by splitting up the region around the vehicle into 3D cubes (i.e., voxels). Then, in each voxel, the likelihood of a pedestrian occupying it is calculated by determining the number of frames in a scene that a pedestrian is occupying the cube over the total number of frames. This is known as the Probabilistic Occupancy Grid (POG).

**[0067]** The mathematical notation for the probability of occupancy of one voxel, p $(v_i)$, is:

$$\hat{p}(v_i) = \frac{\Sigma_{j=1}^{n} 1(ped \in v_{1,j})}{n} \tag{6}$$

**[0068]** Where 1 (•) is the indicator function, $v_i$ is the voxel under consideration, there are n frames of data to consider, and $v_{ij}$ is the voxel state on frame j.

**[0069]** A safety occupancy of pedestrians is calculated in a comparable manner to the general occupancy. The difference is that there are specifications placed alongside the presence of a pedestrian. A specification is a perception awareness attribute to which a system and method for optimization of vehicle perception configuration in accordance with aspects of the present disclosure must adhere. Some non-limiting examples of specifications include: a pedestrian on the sidewalk being of less critical perception nature than a pedestrian walking on the street; a car cutting through traffic from two lanes on the right behind the autonomous vehicle and in a direction towards the front left of the autonomous vehicle being of higher criticality perception then a car which simply changes lanes; detecting a correct right order of arrival at an intersection; and a vehicle passing a red traffic light as opposed to a vehicle properly stopping the red traffic light.

**[0070]** The non-limiting example specifications discussed above are not about algorithms that detect these events, but rather sensor configuration being able to collect enough data, and pass the collected data to a processor that is configured to executed instructions to cause a system for optimization of vehicle perception configuration in accordance with aspects of the present disclosure to properly classify important safety events.

**[0071]** In some embodiments, the specifications may be explicitly listed such that a processor that is configured to executed instructions to cause a system for optimization of vehicle perception configuration in accordance with aspects of the present disclosure may properly classify important safety events. In some embodiments, data-driven methods and machine learning may be implemented to classify each detection as important or not. Then the optimization would be geared toward detecting critical perception events.

**[0072]** In an example embodiment, it is determined whether the pedestrian is also in the street or if the pedestrian is directly in front of the vehicle. These specifications may be set based on the requirements of the system designer and some dependence may need to be considered for the probability distribution to have valid assumptions regarding independence.

**[0073]** A system and method for optimization of vehicle perception configuration in accordance with aspects of the present disclosure will now be described in greater detail with reference to FIGS. **1A**-**10**.

**[0074]** FIG. **1**A illustrates an autonomous vehicle **100** with a sensor configuration thereon.

**[0075]** As shown in the figure, autonomous vehicle **100** includes a forward facing LiDAR **102**, a rearward facing LiDAR **104**, a forward facing RADAR **106**, a rearward facing RADAR **108**, a forward facing camera **110**, a rearward facing camera **112**, a passenger-side facing camera **114**, and a driver-side facing camera **116**.

**[0076]** Forward facing LiDAR **102** is positioned in the center of the front of autonomous vehicle **100** and has a field of view (FOV as shown by a cone) **118**. Rearward facing LiDAR **104** is positioned in the center of the rear of autonomous vehicle **100** and has a FOV as shown by cone **120**. Forward facing RADAR **106** is positioned in the center of the front of autonomous vehicle **100** and has a FOV (as shown by a cone) **122**. Rearward facing RADAR **108** is positioned in the center of the rear of autonomous vehicle **100** and has a FOV as shown by cone **124**. Forward facing camera **110** is positioned in the center of the front of autonomous vehicle **100** and has a FOV (as shown by a cone) **126**. Rearward facing camera **112** is positioned in the center of the rear of autonomous vehicle **100** and has a FOV as shown by cone **128**. Passenger-side facing camera **114** is positioned in the center of the passenger side of autonomous vehicle **100** and has a FOV (as shown by a cone) **130**. Driver-side facing camera **116** is positioned in the center of the driver side of autonomous vehicle **100** and has a FOV as shown by cone **132**.

**[0077]** Each of the FOV's discussed above and illustrated in FIG. **1**A are not drawn to scale and are merely provided to illustrate that different types of perception sensors, e.g., RADAR, LiDAR, camera, have different FOV's and distances, and further that these different perception sensors may be positioned such that multiple FOV's may overlap in areas around autonomous vehicle **100**. For example, in this case, FOV **118** of forward facing LiDAR **102** is wholly overlapped by FOV **122** of forward facing RADAR **106**, which is wholly overlapped by FOV **126** of forward facing camera **110**.

**[0078]** The purpose of the plurality of perception sensors around autonomous vehicle **100** is to perceive obstacles, e.g., pedestrians, around/near autonomous vehicle **100**. In this example, pedestrian **101** may be detected by autonomous vehicle **100** as pedestrian **101** is located within FOV **130** of passenger-side camera **114** and FOV **126** of forward facing camera **110**.

**[0079]** However, the location, orientation, and sensor amount, e.g., in this case the FOV and distance, of each perception sensor is a crucial design aspect for autonomous vehicle **100**. In particular, for example in this instance, there are areas around and near autonomous vehicle **100** that the plurality of perception sensors cannot perceive, such as for example blind spot areas **134**, **136**, **138**, and **140**. This will be further described with reference to FIG. **1**B.

**[0080]** FIG. **1**B illustrates autonomous vehicle **100** with a second sensor configuration thereon. For autonomous vehicle **100** of FIG. **1**B, the sensor configuration is different from that as described above with reference to FIG. **1**A.

**[0081]** As shown in FIG. **1**B, autonomous vehicle **100** includes a forward facing LiDAR **142**, a rearward facing LiDAR **144**, a forward facing RADAR **146**, a rearward facing RADAR **148**, forward facing camera **110**, rearward facing camera **112**, passenger-side facing camera **114**, driver-side facing camera **116**.

**[0082]** Forward facing LiDAR **142** is positioned in the center of the front of autonomous vehicle **100** and has a FOV (as shown by a cone) **150**. Rearward facing LiDAR **144** is positioned in the center of the rear of autonomous vehicle **100** and has a FOV as shown by cone **152**. Forward facing RADAR **146** is positioned in the center of the front of autonomous vehicle **100** and has a FOV (as shown by a cone) **154**. Rearward facing RADAR **148** is positioned in the center of the rear of autonomous vehicle **100** and has a FOV as shown by cone **156**. Forward facing camera **110** is now positioned in the passenger side front quarter panel of autonomous vehicle **100** and has a FOV as shown by cone **158**. Rearward facing camera **112** is now positioned in the driver side rear quarter panel of autonomous vehicle **100** and has a FOV as shown by cone **160**. Passenger-side facing camera **114** is positioned in the passenger side rear quarter panel of autonomous vehicle **100** and has a FOV as shown by cone **162**. Driver-side facing camera **116** is positioned in the driver side front quarter panel of autonomous vehicle **100** and has a FOV as shown by cone **164**.

**[0083]** Each of the FOV's discussed above and illustrated in FIG. **1**B are not drawn to scale and are merely provided to illustrate that different types of perception sensors, e.g., RADAR, LiDAR, camera, have different FOV's and distances, and further that these different perception sensors may be positioned such that multiple FOV's may overlap in areas around autonomous vehicle **100**. For example, in this case, FOV **150** of forward facing LiDAR **142** is wholly overlapped by FOV **154** of forward facing RADAR **146**.

**[0084]** In this example sensor configuration, there are areas around and near autonomous vehicle **100** that the plurality of perception sensors cannot perceive, such as for example blind spot areas **166**, **168**, **170**, **172**, **174**, and **176**.

**[0085]** It should be noted that autonomous vehicle **100** is illustrated as a car, as a non-limiting example provided for discussion purposes. It should be noted that a system and method of perception sensor configuration in accordance with aspects of the present disclosure may be implemented in any type of autonomous vehicle, including aerial and sea-based autonomous vehicles.

**[0086]** FIG. **2**A-B illustrates an example method **200** for optimization of a perception sensor configuration in accordance with aspects of the present disclosure, wherein FIG. **2**A illustrates a first portion of method **200** and FIG. **2**B illustrates a second portion of method **200**.

**[0087]** As shown in FIG. **2**A, method **200** starts (S**202**) and benchmark data is received (S**204**). This will be described in greater detail with reference to FIG. **3**A.

**[0088]** FIG. **3**A illustrates an example perception sensor configuration optimization system **300** in accordance with aspects of the present disclosure at a time $t_3$.

**[0089]** As shown in the figure, perception sensor configuration optimization system **300** includes a plurality of perception sensors **302**, a ground truth simulator system **304**, a benchmark system **306**, a perception sensor configuration optimization system (PCOS) **308**, and a display **310**.

**[0090]** PCOS **308** is configured to: communicate with plurality of perception sensors **302** via a communication channel **312**; communicate with ground truth simulator system **304** via a communication channel **314**; communicate with benchmark system **306** via a communication channel **316**; and communicate with display **310** via a communication channel **318**.

**[0091]** In this example, plurality of perception sensors **302**, ground truth simulator system **304**, benchmark system **306**, PCOS **308**, and display **310** are illustrated as individual devices. However, in some embodiments, at least two of plurality of perception sensors **302**, ground truth simulator system **304**, benchmark system **306**, PCOS **308**, and display **310** may be combined as a unitary device. Further, in some embodiments, at least one of plurality of perception sensors **302**, ground truth simulator system **304**, benchmark system **306**, and PCOS **308** may be implemented as a computer having tangible computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such non-transitory computer-readable recording medium refers to any computer program product, apparatus or device, such as a magnetic disk, optical disk, solid-state storage device, memory, programmable logic devices (PLDs), DRAM, RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired computer-readable program code in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Disk or disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc. Combinations of the above are also included within the scope of computer-readable media. For information transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computer, the computer may properly view the connection as a computer-readable medium. Thus, any such connection may be properly termed a computer-readable medium. Combinations of the above should also be included within the scope of computer-readable media.

**[0092]** Example tangible computer-readable media may be coupled to a processor such that the processor may read information from and write information to the tangible computer-readable media. In the alternative, the tangible computer-readable media may be integral to the processor. The processor and the tangible computer-readable media may reside in an integrated circuit (IC), an application specific integrated circuit (ASIC), or large-scale integrated circuit (LSI), system LSI, super LSI, or ultra LSI components that perform a part or all of the functions described herein. In the alternative, the processor and the tangible computer-readable media may reside as discrete components.

**[0093]** Example tangible computer-readable media may also be coupled to systems, non-limiting examples of which include a computer system/server, which is operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system/server include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, handheld or laptop devices, multiprocessor systems, microprocessor-based systems, set-top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

**[0094]** Such a computer system/server may be described in the general context of computer system-executable instructions, such as program modules, being executed by a computer system. Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types. Further, such a computer system/server may be practiced in distributed cloud computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

**[0095]** Components of an example computer system/server may include, but are not limited to, one or more processors or processing units, a system memory, and a bus that couples various system components including the system memory to the processor.

**[0096]** The bus represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnects (PCI) bus.

**[0097]** A program/utility, having a set (at least one) of program modules, may be stored in the memory by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. The program modules generally carry out the functions and/or methodologies of various embodiments of the application as described herein.

**[0098]** Plurality of perception sensors **302** may include any number of any type of known perception sensors that are configured to perceive the environment within an area and provide distance information input data of obstacles within the area. Non-limiting examples of perception sensors include solid-state LiDAR sensors, mechanical LiDAR sensors, RADAR sensors, ultrasonic sensors, sonar sensors, thermal sensors, and cameras.

**[0099]** Ground truth simulator system **304** may be any known device or system that is configured to simulate, extract scenario data, and generate ground truth information input data **322**. In a non-limiting example, CARLA is implemented as ground truth simulator system **304**. Ground truth information input data **322** is "perfect" safety information-gain. Ground truth information input data **322** is extracted as the data is generated by ground truth simulator system **304**. Ground truth information input data **322** may take the file format as needed or best to represent the data (e.g., spreadsheet, JSON, XML). Perfect detection does not mean the perfect detection only in the FOV. On the contrary, perfect detection may be be all data in the simulation world. What is or is not in the FOV of any particular sensor may be determined by PCOS **308** a later point, as will be described in greater detail below. Ground truth information input data **322** is every piece of relevant information, where acquiring it perfectly could be similar to asking an oracle (a flawless all-knowing thing). In other words, ground truth information input data **322** is not only about objects in the FOV. It should include all objects, so a determination can be made as to whether all objects will be detectable and tractable by the configuration of the plurality of perception sensors **302**.

**[0100]** Benchmark system **306** may be any known device or system that is configured to provide benchmark data **324**. Benchmark data includes data that may be derived from a known standard, e.g., UL 4600, or some organization, and corresponds to predetermined thresholds of acceptability.

**[0101]** PCOS **308**, as will be described in greater detail below, generates a safety occupancy entropy, a general occupancy entropy, and a safety-aware occupancy grid, in accordance with aspects of the present disclosure.

**[0102]** Display **310**, as will be described in greater detail below, may be any known device or system that is configured generate to display a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy.

**[0103]** Communication channels **312**, **314**, **316**, and **318** may be any known type of communication channel, non-limiting examples of which include wired communication channels or wireless communication channels.

**[0104]** In operation, PCOS **308** receives benchmark data from benchmark system **306**. This will be described in greater detail with reference to FIG. **4**A.

**[0105]** FIG. **4**A illustrates an example of PCOS **308** in accordance with aspects of the present disclosure at a time $t_3$.

**[0106]** As shown in the figure, PCOS **308** includes a system controller **402**, a memory **404** having data, instructions, and an optimization program **406** stored therein, a user interface (UI) **408**, a sensor data interface **410**, a simulator data interface **412**, a benchmark data interface **414**, and an output system **416**.

**[0107]** System controller **402** is configured to: communicate with memory **404** via a communication channel **417**; communicate with UI **408** via a communication channel **418**; communicate with sensor data interface **410** via a communication channel **420**; communicate with simulator data interface **412** via a communication channel **422**; communicate with benchmark data interface **414** via a communication channel **424**; and communicate with output system **416** via a communication channel **426**.

**[0108]** Sensor data interface **410** is additionally configured to communicate with plurality of perception sensors **302** via communication channel **312**.

**[0109]** Simulator data interface **412** is additionally configured to communicate with ground truth simulator system **304** via communication channel **314**.

**[0110]** Benchmark data interface **414** is additionally configured to communicate with benchmark system **306** via communication channel **316**.

**[0111]** Output system **416** is additionally configured to communicate with display **310** via communication channel **318**.

**[0112]** In this example, system controller **402**, memory **404**, UI **408**, sensor data interface **410**, simulator data interface **412**, benchmark data interface **414**, and output system **416** are illustrated as individual devices. However, in some embodiments, at least two of system controller **402**, memory **404**, UI **408**, sensor data interface **410**, simulator data interface **412**, benchmark data interface **414**, and output system **416** may be combined as a unitary device. Further, in some embodiments, at least one of system controller **402**, memory **404**, UI **408**, sensor data interface **410**, simulator data interface **412**, benchmark data interface **414**, and output system **416** may be implemented as a computer having tangible computer-readable media for carrying or having computer-executable instructions or data structures stored thereon.

**[0113]** System controller **402** may be implemented as a hardware processor such as a microprocessor, a multi-core processor, a single core processor, a field programmable gate array (FPGA), a microcontroller, an application specific integrated circuit (ASIC), a digital signal processor (DSP), or other similar processing device capable of executing any type of instructions, algorithms, or software for controlling the operation and functions of PCOS **308** in accordance with the embodiments described in the present disclosure.

**[0114]** Memory **404**, as will be described in greater detail below, has instructions, including optimization program **406**, stored therein to be executed by system controller **402** causing PCOS **308** to: determine, based on perception sensor input data **320** and the ground truth information input data **322**, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and output a safety-aware occupancy signal **330** based on the safety occupancy.

**[0115]** In some embodiments, as will be described in greater detail below, optimization program **406** has additional instructions, to be executed by system controller **402**, causing PCOS **308** to: determine, based on perception sensor input data **320** and ground truth information input data **322**, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by: splitting a region around the autonomous vehicle into a plurality of voxels; and calculating the respective likelihood of at least one respective obstacle occupying each voxel; determine, based on the determined general occupancy of the at least one obstacle and the determined safety occupancy of the at least one obstacle, a safety-aware occupancy; and output the safety-aware occupancy signal **330** based on the determined safety-aware occupancy.

**[0116]** In some embodiments, as will be described in greater detail below, optimization program **406** has additional instructions, to be executed by system controller **402**, causing PCOS **308** to: output, to display **310**, safety-aware occupancy signal **330** to cause display **310** to display a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle.

**[0117]** In some embodiments, as will be described in greater detail below, optimization program **406** has additional instructions, to be executed by system controller **402**, causing PCOS **308** to compare the safety-aware occupancy with benchmark data **324**.

**[0118]** In some embodiments, as will be described in greater detail below, optimization program **406** has additional instructions, to be executed by system controller **402**, causing PCOS **308** to determine the safety occupancy by: splitting a region around the autonomous vehicle into a plurality of voxels; calculating, for each voxel, the respective likelihood of an obstacle occupation; and establishing constraints to a presence of any obstacle.

**[0119]** Isome embodiments, when the plurality of perception sensors of the autonomous vehicle include a camera, a RADAR and a LiDAR, as will be described in greater detail below, optimization program **406** has additional instructions, to be executed by system controller **402**, causing PCOS **308** to determine the safety occupancy by determining, based on camera, RADAR and LiDAR input data and ground truth information input data **322**, the safety occupancy of the at least one obstacle within the area around the autonomous vehicle.

**[0120]** In some embodiments, as will be described in greater detail below, optimization program **406** has additional instructions, to be executed by system controller **402**, causing PCOS **308** to enable a user to interface with PCOS **308**, via UI **408**, to cause PCOS **308** to modify the perception sensor input data **320**.

**[0121]** UI **408** may be any known device or system that is configured to enable a user to access and control system controller **402**. UI **408** may include one or more layers including a human-machine interface (HMI) machines with physical input hardware such a keyboards, mice, game pads and output hardware such as computer monitors, speakers, and printers. Additional UI layers in UI **408** may interact with one or more human senses, including: tactile UI (touch), visual UI (sight), and auditory UI (sound).

**[0122]** In some embodiments, as will be described in greater detail below, UI **408** may be configured to enable a user to interface with PCOS **308** to cause PCOS **308** to modify perception sensor input data **320**.

**[0123]** Sensor data interface **410** can include one or more connectors, such as RF connectors, or Ethernet connectors, and/or wireless communication circuitry, such as 5G circuitry and one or more antennas and can operate with corresponding protocols to enable receipt of perception sensor input data **320** from plurality of perception sensors **302**.

**[0124]** Simulator data interface **412** can include one or more connectors, such as RF connectors, or Ethernet connectors, and/or wireless communication circuitry, such as 5G circuitry and one or more antennas and can operate with corresponding protocols to enable receipt of ground truth information input data **322** from ground truth simulator system **304**.

**[0125]** Benchmark data interface **414** can include one or more connectors, such as RF connectors, or Ethernet connectors, and/or wireless communication circuitry, such as 5G circuitry and one or more antennas and can operate with corresponding protocols to enable receipt of benchmark data **324** from benchmark system **306**.

**[0126]** Output system **416** can include one or more connectors, such as RF connectors, or Ethernet connectors, and/or wireless communication circuitry, such as 5G circuitry and one or more antennas and can operate with corresponding protocols to enable output of safety-aware occupancy signal **330** to display **310**.

**[0127]** Communication channels **417**, **418**, **420**, **422**, **424**, and **426** may be any known type of communication channel, non-limiting examples of which include wired communication channels or wireless communication channels.

**[0128]** In operation, benchmark data interface **414** is configured to receive benchmark data **324** from benchmark system **306** via communication channel **316** and provide benchmark data **324** to system controller **402** via communication channel **424**. System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to store benchmark data **324** into memory **404** via communication channel **417**.

**[0129]** Returning to FIG. **2**A, after benchmark data is received (S**204**), perception sensor data is received (S**206**). This will be described in greater detail with reference to FIGS. **3**A, **4**A, **5**A-B, and **6**A-B.

**[0130]** FIG. **5**A illustrates a plan view of autonomous vehicle **100** driving through a portion of a city **500** at a time to. As shown in the figure, portion of city **500** includes an intersection **502** of a north-bound lane **504**, a south-bound lane **506**, a west-bound lane **508**, an east-bound lane **510**, a sidewalk **512**, a sidewalk **514**, a sidewalk **516** and a sidewalk **518**.

**[0131]** Sidewalk **512** is bounded by north-bound lane **504** and east-bound lane **510**. Sidewalk **514** is bounded by south-bound lane **506** and east-bound lane **510**. Sidewalk **516** is bounded by south-bound lane **506** and west-bound lane **508**. Sidewalk **518** is bounded by north-bound lane **504** and west-bound lane **508**.

**[0132]** A plurality of obstacles are strewn around the portion city **500**, a sample of which is indicated as a pedestrian **520**, a construction sign **522**, a construction cone **524**, a construction cone **526**, and a pedestrian **528**.

**[0133]** In operation, autonomous vehicle **100** is driven around the city and the plurality of sensors detect obstacles around autonomous vehicle **100** during the drive. For example, at time to, autonomous vehicle **100** is driving in a north-bound velocity indicated by arrow **532**, wherein forward facing LiDAR **102**, rearward facing LiDAR **104**, forward facing RADAR **106**, rearward facing RADAR **108**, forward facing camera **110**, rearward facing camera **112**, passenger-side facing camera **114**, and driver-side facing camera **116** will detect obstacles within their respective field of view. More specifically, each detector may detect an obstacle within a voxel in its field of view.

**[0134]** A voxel is a single sample, or data point, on a regularly spaced, three-dimensional (3D) grid. In the plan view of FIG. **5**A, the voxels are shown as a grid of 2D pixels, a sample of which is indicated as voxel **530**. The scanning of each voxel will be described in greater detail with reference to FIG. **6**A.

**[0135]** FIG. **6**A illustrates a perspective view of autonomous vehicle **100** and the scanning voxel **530** of FIG. **5**A. While the respective fields of view of each of forward facing LiDAR **102**, forward facing RADAR **106** and driver-side facing camera **116** are not shown, each detector is able to detect voxel **530**. For example, forward facing LiDAR **102** is configured to detect whether voxel **530** includes an obstacle as shown by ray trace **602**, forward facing RADAR **106** is configured to detect whether voxel **530** includes an obstacle as shown by ray trace **604**, and driver-side facing camera **116** is configured to detect whether voxel **530** includes an obstacle as shown by ray trace **606**.

**[0136]** As the number of ray traces increase per voxel, the less uncertainty in determining whether an obstacle is within a voxel. Accordingly, a goal in the sensor suite arrangement is to increase the number of ray traces per voxel. This will be described in greater detail with reference to FIGS. **5**B and **6**B.

**[0137]** FIG. **5**B illustrates a plan view of autonomous vehicle **100** driving through the portion of city **500** at a time $t_1$. As shown in the figure, autonomous vehicle **100** has moved a distance Δy as compared to its position in FIG. **5**A, whereas pedestrian **528** has moved a distance Δx as compared to its position in FIG. **5**A.

**[0138]** Forward facing LiDAR **102**, rearward facing LiDAR **104**, forward facing RADAR **106**, rearward facing RADAR **108**, forward facing camera **110**, rearward facing camera **112**, passenger-side facing camera **114**, and driver-side facing camera **116** will continue to detect obstacles within their respective field of view.

**[0139]** FIG. **6**B illustrates a perspective view of autonomous vehicle **100** and the scanning voxel **530** of FIG. **5**B. In this case, voxel **530** is outside of the field of view of forward facing LiDAR **102**. Accordingly, as compared with that of FIG. **6**A at time $t_0$, at time $t_1$, only forward facing RADAR **106** and driver-side facing camera **116** are able to detect voxel **530**. In particular, forward facing RADAR **106** is configured to detect whether voxel **530** includes an obstacle as shown by ray trace **608**, and driver-side facing camera **116** is configured to detect whether voxel **530** includes an obstacle as shown by ray trace **610**.

**[0140]** Autonomous vehicle **100** continues to drive through the city, and the multiple perception sensors continue to accrue sensor data as perception sensor input data **320**. When the autonomous vehicle **100** completes the drive, the perception sensor input data **320** is downloaded to PCOS **308**. This may be performed by any known method, non-limiting examples of which include wireless and wired transfer of data.

**[0141]** For example, as shown in FIG. **3**A, PCOS **308** is configured to receive perception sensor input data **320**. This will be described in greater detail with reference to FIG. **4**A.

**[0142]** In operation, sensor data interface **410** is configured to receive perception sensor input data **320** from plurality of perception sensors **302** via communication channel **312** and provide perception sensor input data **320** to system controller **402** via communication channel **420**. System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to store perception sensor input data **320** into memory **404** via communication channel **417**.

**[0143]** Returning to FIG. **2**A, after perception sensor data is received (S**206**), perception sensor data is configured (S**208**). For example, as shown in FIG. **3**A, PCOS **308** is configured to configure perception sensor input data **320**. This will be described in greater detail with reference to FIG. **4**A.

**[0144]** As shown in FIG. **4**A, UI **408** is configured to enable a user to configure perception sensor input data **320**. System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to configure the stored perception sensor input data **320** in memory **404** based on the input from the user via UI **408**.

**[0145]** For example, a user may interact with UI **408** to input the specific parameters for each perception sensor, non-limiting examples of such parameters include: the measured location on autonomous vehicle **100**; the type of perception sensor; the orientation; the field of view; and the detecting distance.

**[0146]** System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to modify perception sensor configuration so as to modify the perception sensor input data **320** to take into account the user supplied parameters for each perception sensor.

**[0147]** Returning to FIG. **2**A, after perception sensor data is configured (S**208**), ground truth data is received (S**210**). For example, as shown in FIG. **3**A, PCOS **308** is configured to receive ground truth information input data **322** from ground truth simulator system **304**. This will be described in greater detail with reference to FIG. **4**A.

**[0148]** As shown in FIG. **4**A, simulator data interface **412** is configured to receive ground truth information input data **322** from ground truth simulator system **304** via communication channel **314** and provide ground truth information input data **322** to system controller **402** via communication channel **422**. System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to store ground truth information input data **322** into memory **404** via communication channel **417**.

**[0149]** Returning to FIG. **2**A, after ground truth data is received (S**210**), the safety information-gain is found (S**212**). This will be described in greater detail with reference to FIG. **7**.

**[0150]** FIG. **7** illustrates a more detailed view of the process of finding the safety information-gain (S**212**). As shown in the figure, process S**212** starts (S**702**) and a general occupancy is determined (S**704**). For example, as shown in FIG. **4**A, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to determine a general occupancy.

**[0151]** In particular, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to compare the received perception sensor input data **320**, that corresponds to what each of plurality of perception sensors **302** detected for each voxel around autonomous vehicle **100** for each frame of time, while autonomous vehicle drove around the city, with the ground truth input information data **322** as provided by ground-truth simulator system **304**, in accordance with that as discussed by Hu et al., and with reference to equations (1)-(6) discussed above. From this comparison, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to generate a general occupancy of at least one obstacle within an area around autonomous vehicle **100**.

**[0152]** In some examples, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to generate a general occupancy as a **3**D occupancy grid, for example, as discussed in Hu et al, wherein each voxel in the 3D occupancy grid includes an entropy (uncertainty) value that the voxel will be occupied (by any obstacle) within the next n frames, as discussed above with reference to equation 6. As mentioned above, as autonomous vehicle **100** drives around the city, as the number of ray traces from each of the plurality of perception sensors that pass through each voxel in the 3D occupancy grid increases, the uncertainty that an obstacle may be detected decreases. In other words, the confidence that an obstacle may be detected per voxel in increased.

**[0153]** A 3D occupancy grid may be very complicated for a user to analyze. Therefore, in some embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to then generate a general occupancy signal. This will be described in greater detail with reference to FIGS. **3**B and **4**B.

**[0154]** FIG. **3**B illustrates perception sensor configuration optimization system **300** at a time $t_4$. As shown in the figure, PCOS **308** is configured to store general occupancy data in memory **404** and output a general occupancy signal **326**, based on the general occupancy data, to display **310**. This will be described in greater detail with reference to FIG. **4**B.

**[0155]** FIG. **4**B illustrates PCOS **308** at time $t_4$. As shown in the figure, system controller **402** is configured to execute instructions in optimization program **406** to store general occupancy data in memory **404** and to cause output system **416** to transmit general occupancy signal **326** to display **310**. In some embodiments, general occupancy signal **326** will cause display **310** to display an occupancy grid illustrating perception coverage around autonomous vehicle **100** based on the general occupancy.

**[0156]** In some embodiments, general occupancy signal **326** will cause display **310** to display the occupancy grid as an array representing the area around the autonomous vehicle. In some of these embodiments, the array may take the form of a two-dimensional (2D) array of cells representing a 2D area around autonomous vehicle **100**. In some embodiments, the array may take the form of a three-dimensional (3D) array of cells representing a 3D area, or volume, around autonomous vehicle **100**.

**[0157]** In some embodiments, each cell may be illustrated with a respective numeric value representing a corresponding general occupancy entropy score. In some embodiments, each cell may be illustrated with a respective color representing a corresponding general occupancy entropy score. In some embodiments, each cell may be illustrated with a respective scaled gradient representing a corresponding general occupancy entropy score. In some embodiments, each cell may be represented with a combination of at least one of a respective numeric value, color, and gradient, representing a corresponding general occupancy entropy score.

**[0158]** An example embodiment will now be described in greater detail with reference to FIG. **8**.

**[0159]** FIG. **8** illustrates an example general occupancy grid **800** in accordance with aspects of the present disclosure.

**[0160]** General occupancy grid **800** represents a flattened version of a **3**D occupancy grid mentioned above. General occupancy grid **800** includes a plurality of rows of cells **802** and a plurality of columns of cells **804**.

**[0161]** The left-most portion **806** of general occupancy grid **800** corresponds to locations in front of autonomous vehicle **100**. The rear-most portion **808** of general occupancy grid **800** corresponds to locations behind autonomous vehicle **100**. The top-most portion **810** of general occupancy grid **800** corresponds to locations to the right of autonomous vehicle **100**. The bottom-most portion **812** of general occupancy grid **800** corresponds to locations to the left of autonomous vehicle **100**.

**[0162]** Each cell contains a corresponding entropy score, which is calculated based on the definition of entropy in information theory, discussed above with reference to equations (1)-(6). For a current arrangement of plurality of perception sensors **302**, as shown by general occupancy grid **800**, an area **816** surrounding autonomous vehicle **100** has a decreased uncertainty for detecting obstacles-or an increase in confidence in detecting obstacles. Overall, the shape and size of the area surrounding autonomous vehicle **100** that has decreased uncertainty for detecting obstacles is dependent upon the number, type and orientation of the plurality of perception sensors on autonomous vehicle **100**.

**[0163]** While determining a general occupancy is known, in accordance with aspects of the present disclosure, the probability that a given location near the vehicle will have a perception event occur that is of importance to safety is also determined. In particular, a safety occupancy is determined. For example, as shown in FIG. **5**A, while autonomous vehicle **100** may detect pedestrian **520**, the existence of pedestrian **520** may not have an impact on the safety of vehicle **100** as pedestrian **520** is on sidewalk **512** and is not in the traveling path of autonomous vehicle **100**. On the contrary, pedestrian **528** is in the traveling path of autonomous vehicle **100**. As such, pedestrian **528** may have an impact on the safety of autonomous vehicle **100**.

**[0164]** More generally speaking, in accordance with aspects of the present disclosure, additional predetermined constraints may be placed on detected obstacles to determine a safety occupancy. Non-limiting examples of such additional predetermined constraints may include: weather conditions; road conditions; day/night conditions; conditions on location of an obstacle with reference to the autonomous vehicle; conditions on position of an obstacle with reference to the autonomous vehicle; conditions on a velocity of an obstacle with reference to the autonomous vehicle; conditions on a velocity of the autonomous vehicle; and combinations thereof.

**[0165]** Returning to FIG. **7**, after the general occupancy is determined (S**704**), a safety occupancy is determined (S**706**). This will be described in greater detail with reference to FIGS. **3**C and **4**C.

**[0166]** FIG. **3**C illustrates perception sensor configuration optimization system **300** at a time $t_5$. As shown in the figure, PCOS **308** is configured to output a safety occupancy signal **328** to display **310**. This will be described in greater detail with reference to FIG. **4**C.

**[0167]** FIG. **4**C illustrates PCOS **308** at time $t_5$. As shown in the figure, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to determine a safety occupancy.

**[0168]** In particular, optimization program **406** may have stored therein, predetermined constraints to be placed on detected obstacles to determine a safety occupancy, as noted above. System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to apply the predetermined constraints from optimization program **406** to detected obstacles from received perception sensor input data **320**. In an example embodiment, a predetermined constraint is whether an obstacle is pedestrian and is also in the street or if the pedestrian is directly in front of autonomous vehicle **100**.

**[0169]** Then in a manner similar to the determination of the general occupancy discussed above (S**704**), system controller **402** is configured to execute instructions in optimization program **406** to store safety occupancy data in memory **404** and cause system controller **402** to then generate safety occupancy signal **328**, based on the safety occupancy data.

**[0170]** Further, system controller **402** is configured to execute instructions in optimization program **406** to cause output system **416** to transmit safety occupancy signal **328** to display **310**. In some embodiments, safety occupancy signal **328** will cause display **310** to display a safety occupancy grid illustrating perception coverage around autonomous vehicle **100** based on the safety occupancy. In some embodiments, safety occupancy signal **328** will cause display **310** to display the safety occupancy grid as a 2D array representing the area around the autonomous vehicle. In some embodiments, safety occupancy signal **328** will cause display **310** to display the safety occupancy grid as a 3D array of cells representing a 3D area, or volume, around autonomous vehicle **100**.

**[0171]** In some embodiments, each cell may be illustrated with a respective numeric value representing a corresponding safety occupancy entropy score. In some embodiments, each cell may be illustrated with a respective color representing a corresponding safety occupancy entropy score. In some embodiments, each cell may be illustrated with a respective scaled gradient representing a corresponding safety occupancy entropy score. In some embodiments, each cell may be represented with a combination of at least one of a respective numeric value, color, and gradient, representing a corresponding safety occupancy entropy score.

**[0172]** An example embodiment will now be described in greater detail with reference to FIG. **9**.

**[0173]** FIG. **9** illustrates an example safety occupancy grid **900** in accordance with aspects of the present disclosure.

**[0174]** Safety occupancy grid **900** represents a flattened version of a 3D safety occupancy grid mentioned above. Safety occupancy grid **900** includes a plurality of rows of cells **902** and a plurality of columns of cells **904**.

**[0175]** The left-most portion **906** of safety occupancy grid **900** corresponds to locations in front of autonomous vehicle **100**. The rear-most portion **908** of safety occupancy grid **900** corresponds to locations behind autonomous vehicle **100**. The top-most portion **910** of safety occupancy grid **900** corresponds to locations to the right of autonomous vehicle **100**. The bottom-most portion **912** of safety occupancy grid **900** corresponds to locations to the left of autonomous vehicle **100**.

**[0176]** Each cell contains a corresponding entropy score, which is calculated based on the general occupancy grid **800** as modified by the additional predetermined constraints associated with safety occupancy. For a current arrangement of plurality of perception sensors **302**, as shown by safety occupancy grid **900**, an area **916** surrounding autonomous vehicle **100** has a decreased uncertainty for detecting obstacles—or an increase in confidence in detecting obstacles, that may have an impact on the safety of autonomous vehicle **100**. For example, an area **918** of decreased uncertainty may be attributed to pedestrian **528** (as shown for example in FIGS. **5**A-B) within the traveling path of autonomous vehicle **100**, which may have been detected by each of forward facing LiDAR **102**, forward facing RADAR **106** and forward facing camera **110** (as shown in FIG. **1**A) over a plurality of frames. In some embodiments a weighting factor may be placed on detected obstacles, wherein obstacles associated with safety occupancy have a higher weighting factor than obstacles that are not associated with safety occupancy.

**[0177]** When comparing general occupancy grid **800** as shown in FIG. **8** with safety occupancy grid **900** as shown in FIG. **9**, a general observation is that the overall width of the negative uncertainty cells (cells with higher confidence that an obstacle may be detected) in the safety occupancy grid **900** is much smaller than the general occupancy grid **800**. This is a result of the additional predetermined constraints associated with an impact on the safety of autonomous vehicle **100**. In short, in this example, there is less concern for safety of autonomous vehicle **100** for obstacles to the right and left of autonomous vehicle **100**.

**[0178]** Overall, the shape and size of the area surrounding autonomous vehicle **100** that has decreased uncertainty for detecting obstacles that may have an impact on the safety of autonomous vehicle **100** is dependent upon the number, type and orientation of the plurality of perception sensors on autonomous vehicle **100**.

**[0179]** Returning to FIG. **7**, after the safety occupancy is determined (S**706**), a safety-aware occupancy is determined (S**708**). This will be described in greater detail with reference to FIGS. **3**D and **4**D.

**[0180]** FIG. **3**D illustrates perception sensor configuration optimization system **300** at a time to. As shown in the figure, PCOS **308** is configured to output a safety-aware occupancy signal **330** to display **310**. This will be described in greater detail with reference to FIG. **4**D.

**[0181]** FIG. **4**D illustrates PCOS **308** at time to. As shown in the figure, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to determine a safety-aware occupancy.

**[0182]** In particular, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to subtract the uncertainty value for each cell in the general occupancy grid **800** from the uncertainty value for each corresponding cell in the safety occupancy grid **900** to create a safety-aware occupancy grid.

**[0183]** Further, system controller **402** is configured to execute instructions in optimization program **406** to cause output system **416** to transmit safety-aware occupancy signal **332** to display **310**. In some embodiments, safety-aware occupancy signal **332** will cause display **310** to display the safety-aware occupancy grid illustrating the information that is gained from the safety occupancy constraints on the perception coverage around autonomous vehicle **100**. In some embodiments, safety-aware occupancy signal **332** will cause display **310** to display the safety-aware occupancy grid as a 2D array representing the area around the autonomous vehicle. In some embodiments, safety-aware occupancy signal **332** will cause display **310** to display the safety-aware occupancy grid as a 3D array of cells representing a 3D area, or volume, around autonomous vehicle **100**.

**[0184]** In some embodiments, each cell may be illustrated with a respective numeric value representing a corresponding safety-aware occupancy entropy score. In some embodiments, each cell may be illustrated with a respective color representing a corresponding safety-aware occupancy entropy score. In some embodiments, each cell may be illustrated with a respective scaled gradient representing a corresponding safety-aware occupancy entropy score. In some embodiments, each cell may be represented with a combination of at least one of a respective numeric value, color, and gradient, representing a corresponding safety-aware occupancy entropy score.

**[0185]** An example embodiment will now be described in greater detail with reference to FIG. **10**.

**[0186]** FIG. **10** illustrates an example safety-aware occupancy grid **1000** in accordance with aspects of the present disclosure.

**[0187]** Safety-aware occupancy grid **1000** represents a flattened version of a 3D safety-aware occupancy grid. Safety-aware occupancy grid **1000** includes a plurality of rows of cells **1002** and a plurality of columns of cells **1004**.

**[0188]** The left-most portion **1006** of safety-aware occupancy grid **1000** corresponds to locations in front of autonomous vehicle **100**. The rear-most portion **1008** of safety-aware occupancy grid **1000** corresponds to locations behind autonomous vehicle **100**. The top-most portion **1010** of safety-aware occupancy grid **1000** corresponds to locations to the right of autonomous vehicle **100**. The bottom-most portion **1012** of safety-aware occupancy grid **1000** corresponds to locations to the left of autonomous vehicle **100**.

**[0189]** Each cell in safety-aware occupancy grid **1000** contains a respective entropy score, which is calculated based on a difference between the entropy score of a corresponding cell in general occupancy grid **800** and the entropy score of a corresponding cell in safety occupancy grid **900**.

**[0190]** As a result, a user may easily visualize the total amount of information that is gained from the safety occupancy constraints on the perception coverage around autonomous vehicle **100**. Therefore, in accordance with aspects of the present disclosure, a user can easily distinguish which suite of perception sensors provides optimal detection with respect to the safety of the autonomous vehicle. As mentioned above, in accordance with one or more embodiments of the present disclosure, a user may easily visualize the total amount of information that is gained via at least one of a respective numeric value, color, and gradient of the cells in safety-aware occupancy grid **1000**. Further, as mentioned above, in some embodiments, a safety-aware occupancy grid may take the form of a 3D grid representing the 3D area, or volume, around autonomous vehicle **100**.

**[0191]** Returning to FIG. **7**, after the safety-aware occupancy is determined, process S**212** stops (S**710**).

**[0192]** Returning to FIG. **2**A, after the safety information-gain is found (S**212**) as continued on FIG. **2**B, it is determined whether the current performance meets predetermined benchmarks (S**214**). For example, as shown in FIG. **4**A, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to compare the safety-aware occupancy with benchmark data **324** to determine whether the safety-aware occupancy meets predetermined benchmarks.

**[0193]** Benchmark data **324** provide a set of scenarios under which autonomous vehicle **100** should act safely. They are useful for data collection. System controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to use benchmark data **324** to create a baseline (the ground truth) and use this to determine whether a particular perception configuration has sufficient coverage of safety-aware information. The various perception configurations use the same baseline as the source of comparison to one another. (e.g., configuration A has higher safety-aware occupancy compared to configuration B when considering information from given benchmark)

**[0194]** Returning to FIG. **2**B, if it is determined that the current performance does not meet predetermined benchmarks (N at S**214**), then the performance is removed (S**216**). For example, as shown in FIG. **4**A, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to remove perception sensor input data **320**, the general occupancy data, the safety occupancy data, and the safety-aware occupancy data from memory **404**.

**[0195]** Returning to FIG. **2**B, after the performance is removed (S**216**) then perception sensor data is again configured for another performance (return to S**210** as shown in FIG. **2**A), and method **200** continues. This process may be performed in a manner as discussed above. Alternatively, if it is determined that the current performance does meets predetermined benchmarks (Y at S**214**), then it is determined whether there are more than n performances so far (S**218**). For example, as shown in FIG. **3**A, PCOS **308** is configured to determine whether there are n performances so far. This will be described in greater detail with reference to FIG. **4**A.

**[0196]** As shown in FIG. **4**D, memory **404** may have an integer value, n, stored therein, wherein n represents a predetermined threshold number of configurations of perception sensors to be tested. In some embodiments, system controller **402** is configured to execute instructions in optimization program **406** to enable UI **408** to enable a user to input the integer value for n into memory **404**. In some embodiments, system controller **402** is configured to execute instructions in optimization program **406** to enable UI **408** to enable a user to change a currently stored integer value for n in memory **404**, to a new integer value n'.

**[0197]** System controller **402** is additionally configured to execute instructions in optimization program **406** to cause system controller **402** to determine how many performances, p, have been performed. This may be performed via any known method, a non-limiting example of which includes an incremental digital counter that increases an integer value of p each time perception sensor input data is configured (S**208**).

**[0198]** System controller **402** is additionally configured to execute instructions in optimization program **406** to cause system controller **402** to compare a current integer value p with the integer value n stored in memory **404**. System controller **402** is additionally configured to execute instructions in optimization program **406** to cause system controller **402** to determine that there are not n performances so far, when $p \leq n$. Further, system controller **402** is additionally configured to execute instructions in optimization program **406** to cause system controller **402** to determine that there are n performances so far, when $p > n$.

**[0199]** Returning to FIG. **2**B, if it is determined that there are n performances so far (Y at S**218**), then it is determined whether the current performance is a top-n performance (S**220**). For example, as shown in FIG. **4**D, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to determine whether the current performance is a top-n performance based on predetermined metrics.

**[0200]** In some embodiments, the predetermined metrics for which top-n performances may be ranked are stored in memory **404**. In some embodiments, system controller **402** is configured to execute instructions in optimization program **406** to enable UI **408** to enable a user to input the predetermined metrics into memory **404**. In some embodiments, system controller **402** is configured to execute instructions in optimization program **406** to enable UI **408** to enable a user to change current predetermined metrics in memory **404** to new predetermined metrics.

**[0201]** In some embodiments, the predetermined metrics include an overall size of the area of unit cells in a safety-aware occupancy grid. In these embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to rank the top-n performances based on an overall size of the area of unit cells in a safety-aware occupancy grid that have a negative uncertainty, for example as shown in area **1014** of FIG. **10**. In these embodiments, the performances that have the largest overall area of safety-aware detection are valued more.

**[0202]** In some embodiments, the predetermined metrics include an average value of uncertainty of the area of unit cells in a safety-aware occupancy grid. In these embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to rank the top-n performances based on an average value of uncertainty of the area of unit cells in a safety-aware occupancy grid that have a negative uncertainty, for example as shown in area **1014** of FIG. **10**. In these embodiments, the performances that have the best average safety-aware detection are valued more.

**[0203]** In some embodiments, the predetermined metrics include the area of unit cells in a predetermined position of the area of unit cells in a safety-aware occupancy grid. In these embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to rank the top-n performances based on the area of unit cells in a predetermined position in a safety-aware occupancy grid that have a negative uncertainty, for example a left-most portion **1006** of safety-aware occupancy grid **1000** that corresponds to locations in front of autonomous vehicle **100** as shown in area **1014** of FIG. **10**. In these embodiments, the performances that have the best safety-aware detection in a particular direction or plural directions are valued more.

**[0204]** In some embodiments, the predetermined metrics include a total sum value of uncertainty of the area of unit cells in a safety-aware occupancy grid. In these embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to rank the top-n performances based on a total sum value of uncertainty of the area of unit cells in a safety-aware occupancy grid that have a negative uncertainty, for example as shown in area **1014** of FIG. **10**. In these embodiments, the performances that have the most precise safety-aware detection are valued more.

**[0205]** In some embodiments, the predetermined metrics include a total monetary cost of the plurality of perception sensors (e.g., spending $X may result in Y % more safety occupancy score). In these embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to rank the top-n performances based on total monetary cost of the plurality of perception sensors. In these embodiments, the performances that have the greatest ratio of occupancy score per total price of sensors are valued more.

**[0206]** In some embodiments, the predetermined metrics include a combination of at least two of the above-discussed predetermined metrics. In these embodiments, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to rank the top-n performances based on a combination of at least two of: an overall size of the area of unit cells in a safety-aware occupancy grid that have a negative uncertainty; an average value of uncertainty of the area of unit cells in a safety-aware occupancy grid that have a negative uncertainty; the area of unit cells in a predetermined position in a safety-aware occupancy grid that have a negative uncertainty; a total monetary cost of the plurality of perception sensors; and a total sum value of uncertainty of the area of unit cells in a safety-aware occupancy grid that have a negative uncertainty.

**[0207]** In operation, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to determine whether the safety-aware occupancy grid of the current performance is superior to one of the safety-aware occupancy grid of other top-n performances based on the predetermined metrics discussed above.

**[0208]** Returning to FIG. **2**B, if it is determined that the current performance is not a top-n performance (N at S**220**), then the performance is removed (S**216**), perception sensor data is again configured for another performance (return to S**210** as shown in FIG. **2**A), and method **200** continues. This process may be performed in a manner as discussed above.

**[0209]** In other words, autonomous vehicle **100** is outfitted with a plurality of perception sensors, each in a respective location and having a respective orientation on autonomous vehicle **100**, and each having a respective field of view and perception capability, for example as shown in FIG. **1**A. After the autonomous vehicle **100** is driven around and perception data is captured, in accordance with aspects of the present disclosure: a general occupancy and corresponding general occupancy grid, such as for example as discussed above with reference to FIG. **8**, are generated; a safety occupancy and corresponding safety occupancy grid, such as for example as discussed above with reference to FIG. **9**, are generated; and a safety-aware occupancy and corresponding safety-aware occupancy grid, such as for example as discussed above with reference to FIG. **10**, are generated.

**[0210]** Then a new plurality of perception sensors are mounted on autonomous vehicle **100**. This "new" plurality of perception sensors may include at least one of the previously mounted perception sensors, yet having been mounted in at least one of a new location on autonomous vehicle **100** or a new orientation on autonomous vehicle, for example as shown in FIG. **1**B. Then again the autonomous vehicle **100** is driven around and new perception data is captured. Then, in accordance with aspects of the present disclosure: a new general occupancy and corresponding new general occupancy grid are generated; a new safety occupancy and corresponding new safety occupancy grid are generated; and a new safety-aware occupancy and corresponding new safety-aware occupancy grid are generated.

**[0211]** Alternatively, if it is determined that the current performance is a top-n performance (Y at S220), then current worst performance of the current top-n performances is removed (S222). For example, as shown in FIG. **3**A, PCOS **308** is configured to remove the current worst performance of the current top-n performances. This will be described in greater detail with reference to FIG. **4**D.

**[0212]** As shown in FIG. **4**D, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to remove perception sensor input data of the worst performance, the general occupancy data of the worst performance, the safety occupancy data of the worst performance, and the safety-aware occupancy data of the worst performance from memory **404**.

**[0213]** Returning to FIG. **2**B, after the current worst performance of the current top-n performances is removed (S**222**), then the current performance is added to the top-n performances (S**224**). For example, as shown in FIG. **3**A, PCOS **308** is configured to add the current performance to the top-n performances. This will be described in greater detail with reference to FIG. **4**D.

**[0214]** As shown in FIG. **4**A, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to maintain perception sensor input data of the current performance, the general occupancy data of the current performance, the safety occupancy data of the current performance, and the safety-aware occupancy data of the current performance in memory **404**.

**[0215]** Returning to FIG. **2**B, alternatively, if it is determined that there are not more than n performances so far (N at S**218**), then the current performance is added to the top-n performances (S**224**). This may be performed in a manner as discussed above. After the current performance has been added to the top-n performances (S**224**), then it is determined whether to terminate performances (S**226**). For example, as shown in FIG. **3**A, PCOS **308** is configured to determine whether to terminate performances. This will be described in greater detail with reference to FIG. **4**D.

**[0216]** As shown in FIG. **4**D, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to terminate performances. For example, in some embodiments, criteria for stopping may be stored in memory **404**, wherein system controller **402** is configured to execute instructions in optimization program **406** to obtain the stopping criteria from memory **404**. Non-limiting examples of stopping criteria include, a predetermined fixed number of performances, a predetermined convergence value, a predetermined time of performing performances, and combinations thereof.

**[0217]** Further, in some embodiments, a user may use UI **408** to input, modify, or change the criteria for stopping in memory **404**, wherein system controller **402** is configured to execute instructions in optimization program **406** to, based on input from UI **408**, input, modify, or change the criteria for stopping in memory **404**.

**[0218]** Returning to FIG. **2**B, if it is determined not to terminate performances (N at S**226**), then perception sensor data is again configured for another performance (return to S**210** as shown in FIG. **2**A), and method **200** continues. This may be performed in a manner as discussed above. Alternatively, if it is determined to terminate performances (Y at S**226**), then the performances are outputted (S**228**). This will be described in greater detail with reference to FIG. **3**E.

**[0219]** FIG. **3**E illustrates perception sensor configuration optimization system **300** at a time $t_7$. As shown in the figure, PCOS **308** is configured to output the performances. This will be described in greater detail with reference to FIG. **4**E.

**[0220]** FIG. **4**E illustrates PCOS **308** at a time $t_7$. As shown in the figure, system controller **402** is configured to execute instructions in optimization program **406** to cause system controller **402** to cause display **310** to display the top-n safety-aware occupancy grids. In this manner, a user can easily visualize and compare the top-n safety-aware occupancy grids. In particular, a user may easily visually evaluate aspects of the top-n safety-aware occupancy grids with respect to at least one of: an overall area of each grid, thus relating to the overall increased information that is gained from the safety occupancy constraints on the perception coverage around autonomous vehicle **100** for each performance, i.e., each sensor suite arrangement; a particular area of interest of each grid, thus relating to an increased information that is gained from the safety occupancy constraints on the perception coverage around autonomous vehicle **100** for each performance, i.e., each sensor suite arrangement, for a particular area, such as the front of autonomous vehicle **100**;. the monetary cost of the sensor configuration; and the return on investment of one sensor configuration over another.

**[0221]** By visualizing the different top-n safety-aware occupancy grids, a user may determine which perception sensor suite arrangement is most beneficial. It should be noted that the "most beneficial" perception sensor suite arrangement might not be the "best" perception sensor suite arrangement based on judging criteria mentioned above. On the contrary, a user may determine that the "best" perception sensor suite arrangement based on judging criteria mentioned above might be too costly to implement via re-tooling a manufacturing line or purchasing the needed perception sensors. In this light, a user may alternatively determine that the "most beneficial" perception sensor suite is not the "best" perception sensor suite arrangement based on judging criteria mentioned above, but in consideration of implementation in a current manufacturing line, the "most beneficial" perception sensor suite is the $5^{th}$ sensor suite corresponding to the $5^{th}$ safety-aware occupancy grid of provided the top-n safety-aware occupancy grids.

**[0222]** Returning to FIG. **2**B, after the performances are outputted (S**228**), method **200** stops (S**230**).

**[0223]** The terminology used herein is for the purpose of describing particular implementations only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, including any steps performed by a/the computer/processor, unless the context clearly indicates otherwise. As used herein, the phrase "at least one of A, B, and C" should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean "at least one of A, at least one of B, and at least one of C." As another example, the language "at least one of A and B" (and the like) as well as "at least one of A or B" (and the like) should be interpreted as covering only A, only B, or both A and B, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps (not necessarily in a particular order), operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps (not necessarily in a particular order), operations, elements, components, and/or groups thereof. Example sizes/models/values/ranges can have been given, although examples are not limited to the same.

**[0224]** The terms (and those similar to) "coupled," "attached," "connected," "adjoining," "transmitting," "receiving," "connected," "engaged," "adjacent," "next to," "on top of," "above," "below," "abutting," and "disposed," used herein is to refer to any type of relationship, direct or indirect, between the components in question, and is to apply to electrical, mechanical, fluid, optical, electromagnetic, electromechanical, or other connections. Additionally, the terms "first," "second," etc. are used herein only to facilitate discussion, and carry no particular temporal or chronological significance unless otherwise indicated. The terms "cause" or "causing" means to make, force, compel, direct, command, instruct, and/or enable an event or action to occur or at least be in a state where such event or action is to occur, either in a direct or indirect manner. The term "set" does not necessarily exclude the empty set-in other words, in some circumstances a "set" may have zero elements. The term "non-empty set" may be used to indicate exclusion of the empty set-that is, a non-empty set must have one or more elements, but this term need not be specifically used. The term "subset" does not necessarily require a proper subset. In other words, a "subset" of a first set may be coextensive with (equal to) the first set. Further, the term "subset" does not necessarily exclude the empty set-in some circumstances a "subset" may have zero elements.

**[0225]** The corresponding structures, materials, acts, and equivalents (e.g., of all means or step plus function elements) that may be in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. While the disclosure describes structures corresponding to claimed elements, those elements do not necessarily invoke a means plus function interpretation unless they explicitly use the signifier "means for." Unless otherwise indicated, recitations of ranges of values are merely intended to serve as a shorthand way of referring individually to each separate value falling within the range, and each separate value is hereby incorporated into the specification as if it were individually recited. While the drawings divide elements of the disclosure into different functional blocks or action blocks, these divisions are for illustration only. According to the principles of the present disclosure, functionality can be combined in other ways such that some or all functionality from multiple separately-depicted blocks can be implemented in a single functional block; similarly, functionality depicted in a single block may be separated into multiple blocks. Unless explicitly stated as mutually exclusive, features depicted in different drawings can be combined consistent with the principles of the present disclosure. Moreover, although this disclosure describes and depicts respective implementations herein as including particular components, elements, feature, functions, operations, or steps (and arrangements thereof), any of these implementations may include any combination, arrangement, or permutation of any of the components, elements, features, functions, operations, or steps described or depicted anywhere herein that a person having ordinary skill in the art would comprehend after reading the present disclosure. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative.

**[0226]** The description of the present disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the disclosure in the form disclosed. After reading the present disclosure, many modifications, variations, substitutions, and any combinations thereof will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The implementation(s) were chosen and described in order to explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various implementation(s) with various modifications and/or any combinations of implementation(s) as are suited to the particular use contemplated. The features of any dependent claim may be combined with the features of any of the independent claims or other dependent claims.

**[0227]** Having thus described the disclosure of the present application in detail and by reference to implementation(s) thereof, it will be apparent that modifications, variations, and any combinations of implementation(s) (including any modifications, variations, substitutions, and combinations thereof) are possible without departing from the scope of the disclosure defined in the appended claims.

What is claimed is:

**1**. A perception sensor configuration optimization system comprising:
- a memory having instructions stored therein; and
- a processor configured to execute the instructions stored in said memory to cause said perception sensor configuration optimization system to:
  - receive perception sensor input data of an autonomous vehicle, wherein the perception sensor input data corresponds to respective measured location, orientation, and type of a plurality of perception sensors of the autonomous vehicle;
  - receive ground-truth information input data of the autonomous vehicle, wherein the ground-truth information input data corresponds to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle;
  - determine, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and
  - output a safety-aware occupancy signal based on the safety occupancy.

**2**. The perception sensor configuration optimization system of claim **1**, wherein said processor is further configured to execute the instructions stored in said memory to cause said perception sensor configuration optimization system to:
- determine, based on the perception sensor input data and the ground-truth information input data, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by:
  - splitting a region around the autonomous vehicle into a plurality of voxels; and
  - calculating a respective likelihood of the at least one obstacle occupying each voxel of the plurality of voxels;
- determine, based on the general occupancy of the at least one obstacle and the safety occupancy of the at least one obstacle, a safety-aware occupancy; and
- output the safety-aware occupancy signal based on the safety-aware occupancy.

**3**. The perception sensor configuration optimization system of claim **2**, wherein said processor is further configured to execute the instructions stored in said memory to cause said perception sensor configuration optimization system to output, to a display, the safety-aware occupancy signal to display a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle.

**4**. The perception sensor configuration optimization system of claim **2**, wherein said processor is further configured to execute the instructions stored in said memory to additionally cause said perception sensor configuration optimization system to compare the safety-aware occupancy with benchmark data corresponding to predetermined thresholds of acceptability.

**5**. The perception sensor configuration optimization system of claim **1**, wherein said processor is further configured to execute the instructions stored in said memory to cause said perception sensor configuration optimization system to determine the safety occupancy by:
- splitting a region around the autonomous vehicle into a plurality of voxels;
- calculating, for each voxel of the plurality of voxels, a respective likelihood of an obstacle occupation; and
- establishing constraints to a presence of an obstacle.

**6**. The perception sensor configuration optimization system of claim **1**,
- wherein the plurality of perception sensors of the autonomous vehicle include a distance perception sensor configured to provide distance information input data of the at least one obstacle, and
- wherein said processor is further configured to execute the instructions stored in said memory to cause said perception sensor configuration optimization system to determine the safety occupancy by determining, based on the distance information input data and the ground-truth information input data, the safety occupancy of the at least one obstacle within the area around the autonomous vehicle.

**7**. The perception sensor configuration optimization system of claim **1**, further comprising a user interface configured to enable a user to modify the perception sensor input data.

**8**. A method comprising:
- receiving perception sensor input data of an autonomous vehicle, the perception sensor input data corresponding to respective measured location, orientation, and type of each of a plurality of perception sensors of the autonomous vehicle;
- receiving ground-truth information input data of the autonomous vehicle, the ground-truth information input data corresponding to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle;
- determining, via a processor configured to execute instructions stored in a memory, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and
- outputting, via the processor, a safety-aware occupancy signal based on the safety occupancy.

**9**. The method of claim **8**, further comprising:
- determining, via the processor, based on the perception sensor input data and the ground-truth information input data, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by:

splitting, via the processor, a region around the autonomous vehicle into a plurality of voxels; and
calculating, via the processor, a respective likelihood of the at least one obstacle occupying each voxel of the plurality of voxels;
determining, via the processor, based on the general occupancy of the at least one obstacle and the safety occupancy of the at least one obstacle, a safety-aware occupancy; and
outputting, via the processor, the safety-aware occupancy signal based on the safety-aware occupancy.

**10**. The method of claim **9**, further comprising displaying, via a display, a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle.

**11**. The method of claim **9**, further comprising:
receiving, via the processor and from a benchmark system, benchmark data corresponding to predetermined thresholds of acceptability; and
comparing, via the processor, the safety-aware occupancy with the benchmark data.

**12**. The method of claim **8**, wherein said determining the safety occupancy comprises:
splitting, via the processor, a region around the autonomous vehicle into a plurality of voxels;
calculating, via the processor and for each voxel of the plurality of voxels, a respective likelihood of an obstacle occupation; and
establishing, via the processor, constraints to a presence of an obstacle.

**13**. The method of claim **8**, wherein said determining the safety occupancy of obstacles within the area around the autonomous vehicle comprises determining, via the processor and based on distance information input data of the at least one obstacle and the ground-truth information input data, the safety occupancy of obstacles within the area around the autonomous vehicle.

**14**. The method of claim **8**, further comprising enabling, via a user interface, a user to modify the perception sensor input data.

**15**. A non-transitory, computer-readable media having computer-readable instructions stored thereon, which, when executed across one or more processors, causes at least a portion of the one or more processors to perform operations comprising:
receiving perception sensor input data of an autonomous vehicle, the perception sensor input data corresponding to respective measured location, orientation, and type of a plurality of perception sensors of the autonomous vehicle;
receiving ground-truth information input data of the autonomous vehicle, the ground-truth information input data corresponding to respective ideal simulated location, ideal orientation, and ideal type of a plurality of perception sensors of the autonomous vehicle;
determining, via a processor configured to execute instructions stored in a memory, based on the perception sensor input data and the ground-truth information input data, a safety occupancy of at least one obstacle within an area around the autonomous vehicle; and
outputting, via the processor, a safety-aware occupancy signal based on the safety occupancy.

**16**. The non-transitory, computer-readable media of claim **15**, wherein the operations further comprise:
determining, based on the perception sensor input data and the ground-truth information input data, a general occupancy of the at least one obstacle within the area around the autonomous vehicle by:
splitting a region around the autonomous vehicle into a plurality of voxels; and
calculating a respective likelihood of the at least one obstacle occupying each voxel of the plurality of voxels;
determining based on the general occupancy of the at least one obstacle and the safety occupancy of the at least one obstacle, a safety-aware occupancy; and
outputting the safety-aware occupancy signal based on the safety-aware occupancy.

**17**. The non-transitory, computer-readable media claim **16**, wherein the operations further comprise displaying, via a display, a safety-aware occupancy grid illustrating perception coverage around the autonomous vehicle based on the safety-aware occupancy as an array representing the area around the autonomous vehicle.

**18**. The non-transitory, computer-readable media of claim **15**, wherein determining the safety occupancy comprises:
splitting a region around the autonomous vehicle into a plurality of voxels;
calculating, for each voxel of the plurality of voxels, a respective likelihood of an obstacle occupation; and
establishing constraints to a presence of an obstacle.

**19**. The non-transitory, computer-readable media of claim **15**, wherein determining the safety occupancy of obstacles within the area around the autonomous vehicle comprises determining, via distance information input data of the at least one obstacle and the ground-truth information input data, the safety occupancy of obstacles within the area around the autonomous vehicle.

**20**. The non-transitory, computer-readable media of claim **15**, wherein the operations further comprise enabling, via a user interface, a user to modify the perception sensor input data.

* * * * *